

Chemistry

CYL1010



Google Class code: **izdaj53f**

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Contents

Title	Chemistry	Number	CYL1010
Department	Chemistry	L-T-P [C]	3-0-0 [3]
Offered for	B.Tech. 1 st Year	Type	Compulsory
Prerequisite	Nil		

Objectives

1. Relate the fundamental principles of chemistry with practical problems encountered for engineers and place the emphasis on problem-solving
2. Provide knowledge of engineering materials metals, composites, polymers, cement, steel, fuels, batteries and drugs.

Learning Outcomes

1. Understanding the behavior of matter and materials using fundamental knowledge of their nature.
2. Understanding the basic concepts and implementation of the same in advanced and interdisciplinary areas.
3. Predict potential complications from combining various chemicals or metals in an engineering setting.

Contents

Atomic and Molecular Structure [6 Lectures]: particle in a box, angular momentum, wavefunction, multielectron atoms, homonuclear and heteronuclear diatomic molecules, non-covalent interactions

Solid State Chemistry [6 Lectures]: Lattice and Unit Cells, X-Ray Diffraction, Bands in solids, semiconductors, Introduction to nanomaterials based on dimensionality and properties.

Surface Chemistry[3 Lectures]: Surfaces and interfaces, surface modification, surface energy, contact angle and surface tension.

Electrochemical Systems [5 Lectures]: Electrochemical cells and EMF, Applications of EMF measurements: Steady state approximation, Chain reactions, photochemical kinetics, Fuel Cells and Batteries, OER and HER reactions, Corrosion

Basic Spectroscopy [7 Lectures]: Selection Rules, Fundamentals FTIR, Raman , UV-visible and Fluorescence spectroscopy

Transition Metal Chemistry [5 Lectures]: Colors in Complexes, Organometallics and Catalysis, Bioinorganic Chemistry

Organic Chemistry [8 Lectures]: Name Reactions, Stereochemistry and their Applications, Acid-Base Chemistry, Use of Buffers, Organic Chemistry examples such as polymers and biopolymers, drugs and photoactive molecules in everyday life and advanced technologies.

Textbooks

1. Silberberg, M., *Chemistry: The Molecular Nature of Matter and Change*, 6th Edition, McGraw Hill Education.
2. West, A.R., (2015), *Solid State Chemistry and Its Applications*, 2nd edition, John Wiley & Sons.
3. J. D. Lee, Concise Inorganic Chemistry, 5th Edition.
4. Atkins, Physical Chemistry, 11th Edition.

Reference Books

1. McMurry, J. E. & Fay, R. C. *Chemistry*, 5th Edition, Pearson.
2. Jonathan Clayden, Nick Greeves, Stuart Warren, Organic Chemistry, 2nd Edition.
3. F. Albert Cotton, Geoffrey Wilkinson, Basic Inorganic Chemistry, Wiley Student Edition.

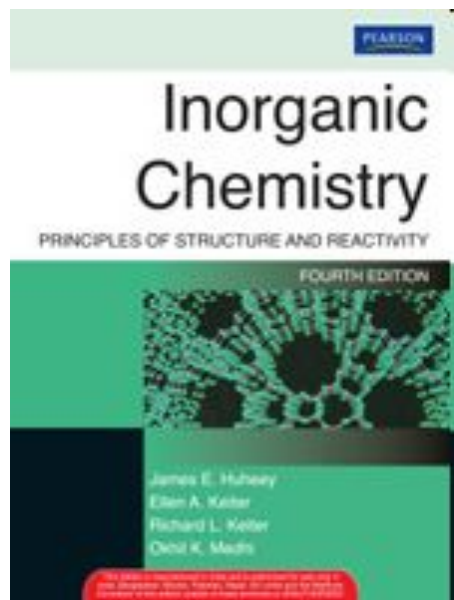
Self Learning Material

1. <https://nptel.ac.in/downloads/122101001/>

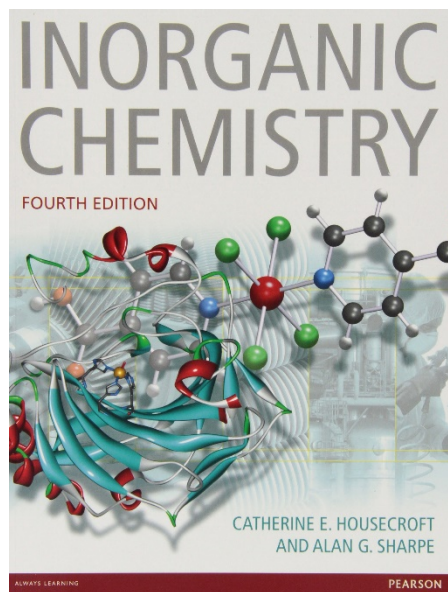
Contents

- **Solid State Chemistry [6 Lectures]:** Lattice and Unit Cells, X-Ray Diffraction, Bands in solids, semiconductors, Introduction to nanomaterials based on dimensionality and properties.
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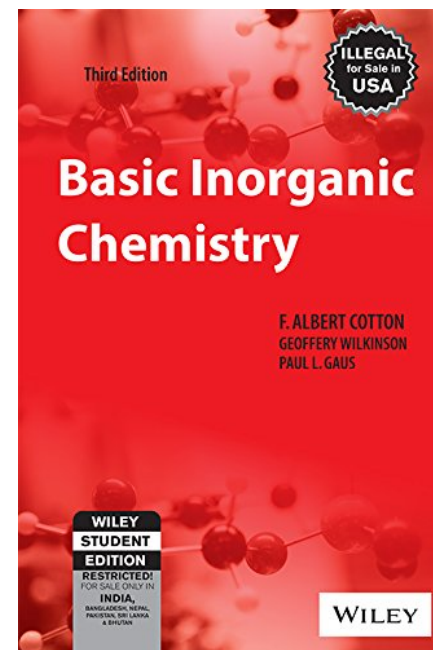
Reference Books



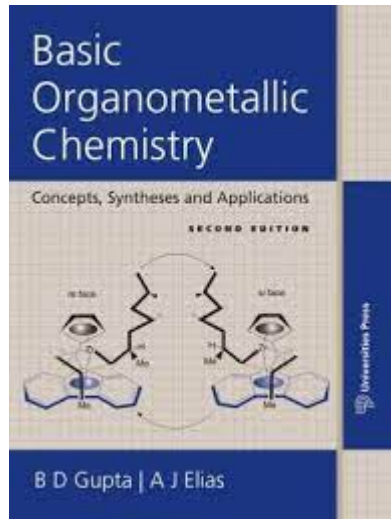
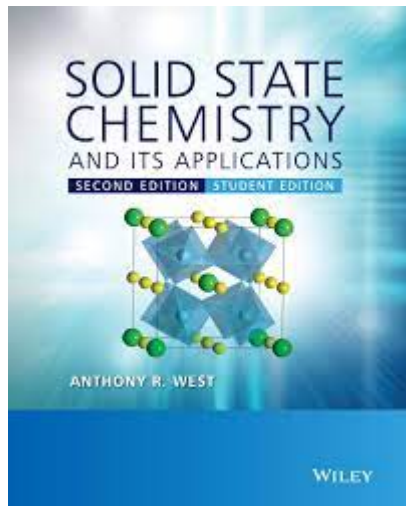
J. E. Huheey
Keiter & Keiter



Catherine E. Housecroft
and Alan G. Sharpe



F. Albert Cotton,
Geoffrey Wilkinson



Course Grading Scheme

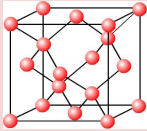
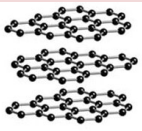
Quiz 1	10%
Quiz 2	10%
Assignment	5 %
Interaction	5%
Minor	30%
Major	40%
TOTAL	100 %

Why do you study the structure of crystalline solids?

□ The properties of some materials are directly related to their crystal structures.

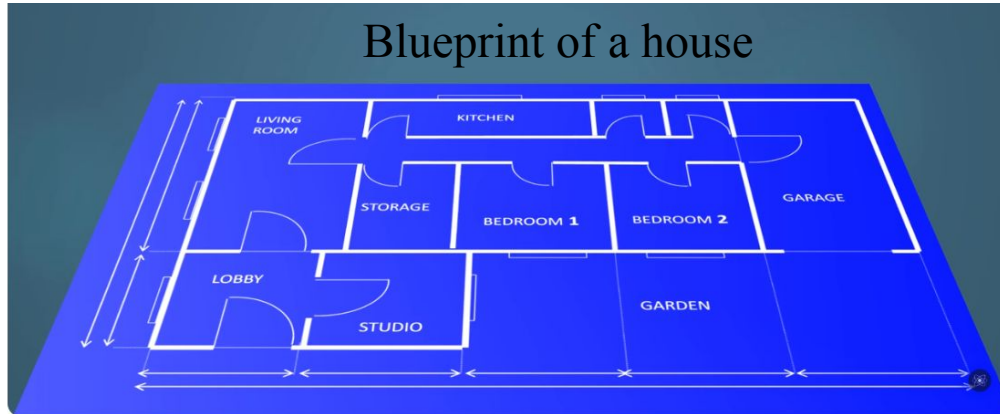
For example, pure and undeformed magnesium and beryllium, having one crystal structure, are much more brittle (i.e., fracture at lower degrees of deformation) than are pure and undeformed metals such as gold and silver that have yet another crystal structure.

□ significant property differences exist between crystalline and noncrystalline materials having the same composition. For example, noncrystalline ceramics and polymers normally are optically transparent; the same materials in crystalline (or semicrystalline) form tend to be opaque or, at best, translucent

C	Diamond	Graphite
carbon		
	transparent, hard, insulating, expensive	black, soft, conducting, cheap
		

Lattice , Basis and Crystal

Blueprint of a house



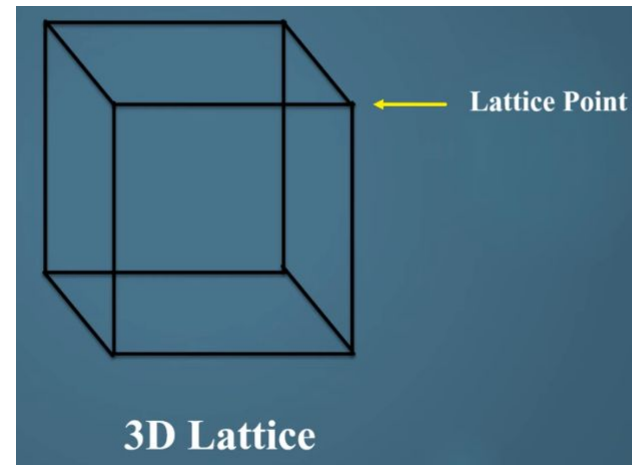
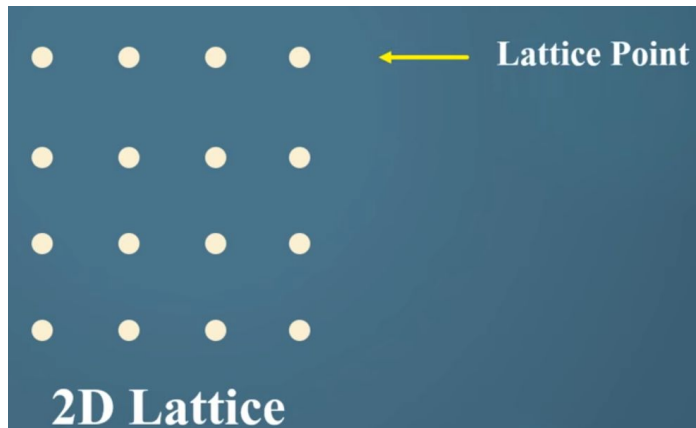
Real house after building up the walls



Fundamental terms of crystallography

Lattice

- *A lattice is a hypothetical structure with regular and periodic arrangement of points in three dimensions.*

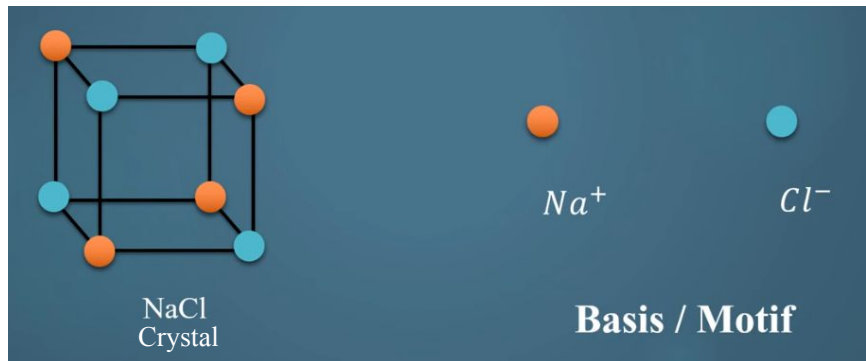


It can be expressed in another way.

- *In an arrangement of points, if the surrounding environment looks like same when the arrangement is viewed from the different lattice points, then that arrangement is said to be lattice.*

Basis

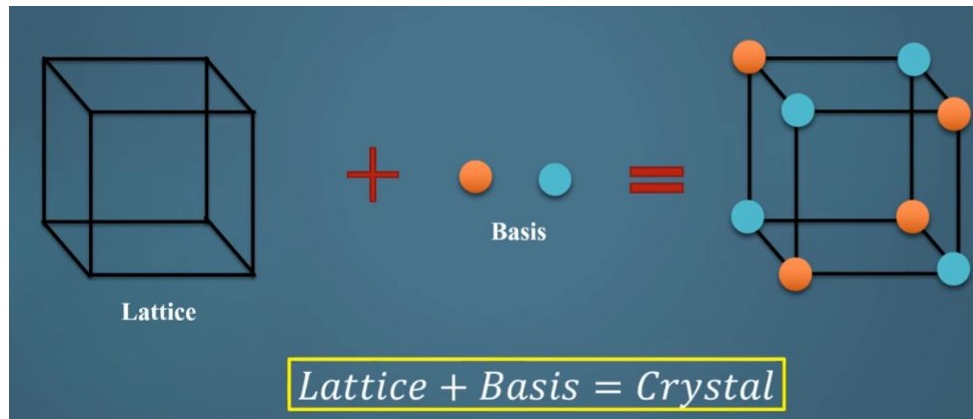
- To construct the crystal structure, some basic arrangement is to be fixed at each and every lattice point.
- The basic arrangement is said to be basis or motif



The lattice of a crystal is a regular, repeating arrangement of points in space. The basis is a set of atoms or molecules associated with each lattice point. When combined with the lattice, the basis forms the crystal structure.

Crystal structure

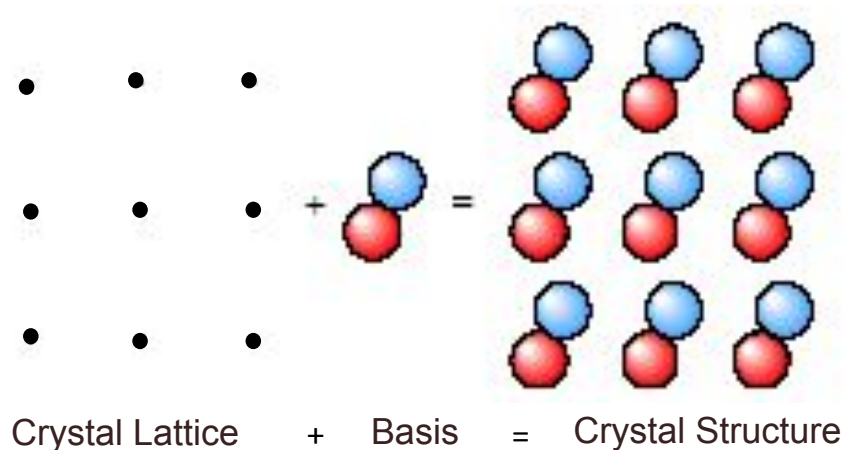
- A crystal structure is obtained by arranging the basis in each and every lattice point. It can be written as
Crystal structure = lattice + basis
- This is used to explain the formation of crystal structure



Crystal Structure

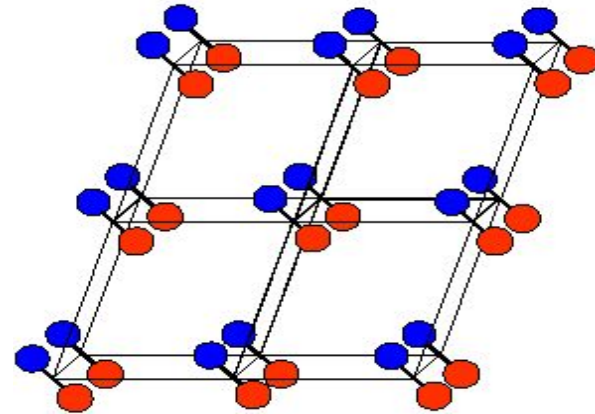
- Crystal structure can be obtained by attaching atoms, groups of atoms or molecules which are called basis (motif) to the lattice sites of the lattice point.

Crystal Structure = Crystal Lattice • + Basis 



Crystal structure

- Don't mix up atoms with lattice points
- Lattice points are infinitesimal points in space
- Lattice points do not necessarily lie at the centre of atoms



Crystal Structure = Crystal Lattice • + Basis 

Important Note:

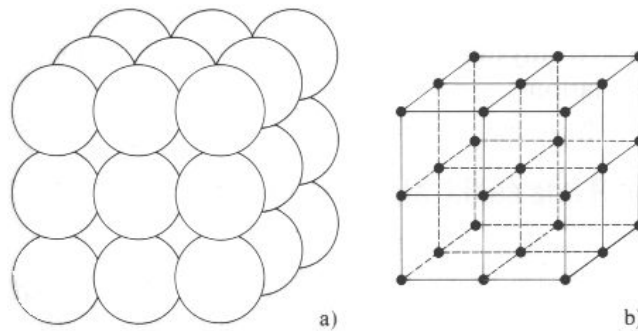
- Lattice points are a purely mathematical concept, whereas atoms are physical objects.

For example, the only element exhibiting Simple Cubic structure is Po (*Polonium*) at ambient conditions.

In Figure (a) is the 3-D periodic arrangement of Po atoms, and

Figure (b) is the corresponding space lattice.

In this case, atoms lie at the same point as the space lattice.



Definition 1

$$\text{Crystal} = \text{Lattice} + \text{Motif}$$

Motif or **Basis**:

an entity (typically an atom or a group of atoms) associated with each lattice point

Lattice □ the underlying periodicity of the crystal

Motif □ Entity **associated** with each lattice point

Lattice □ how to repeat

Motif □ what to repeat

Lattice

Translationally periodic
arrangement of **points**

Crystal

Translationally
periodic arrangement
of **motifs**

Crystal structure of a material

- Some of the important parameters which are used to describe the crystal structure of materials.
 - Atomic radius
 - Coordination number
 - Density of packing

Atomic Radius

- It is the half the distance between the any two successive atoms.

For example cubic system,

$$r = a / 2$$

where a is the interatomic distance

Coordination Number

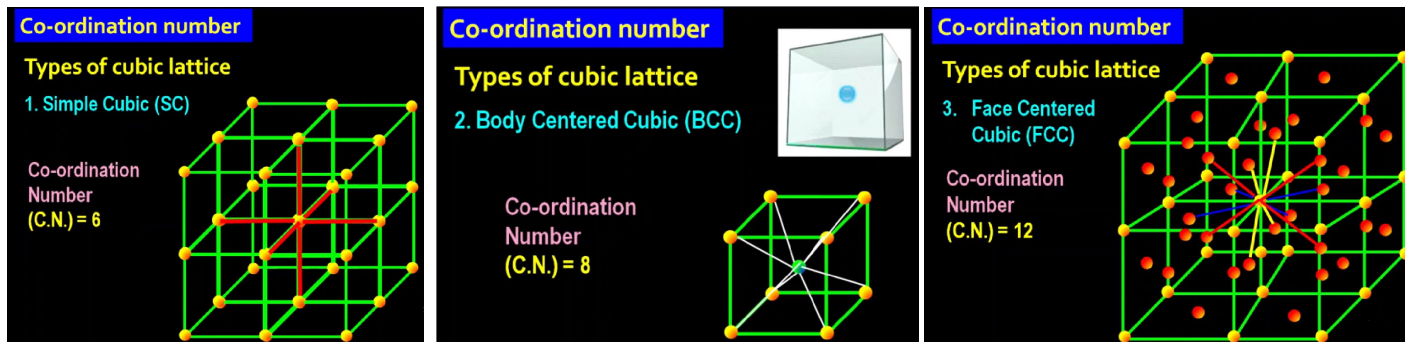
- It is the number of nearest neighboring atoms to a particular atom.
 - e.g.. Simple cubic unit cell, the coordination number is 6

Coordination Number

Coordination Number (CN) : The Bravais lattice points closest to a given point are the nearest neighbours.

Because the Bravais lattice is periodic, all points have the same number of nearest neighbours or coordination number. It is a property of the lattice.

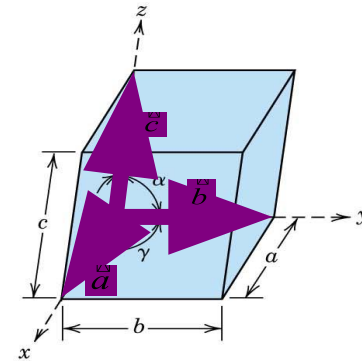
A simple cubic has coordination number 6; a body-centered cubic lattice, 8; and a face-centered cubic lattice 12.



Note: *The only element exhibiting Simple Cubic structure is Po (Polonium) at ambient conditions.*

Primitives

- Let a , b , c are the length along the axes x, y, z in cubic cell is called primitives

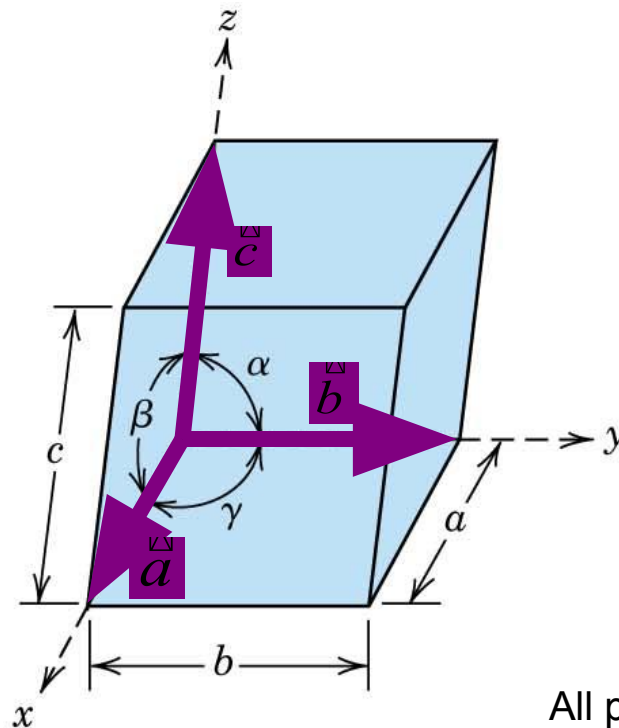


Interaxial angles or interfacial angles

- In the crystal, the angles between x , y and z axes are called interfacial angles it is represented by α , β and γ

Lattice parameter

- To represent a unit cell, the interfacial angles and intercept in axes are important.
- These parameters are called lattice parameters



Lattice parameters

axial lengths: a, b, c

interaxial angles: α, β, γ

unit vectors: $\vec{a} \quad \vec{b} \quad \vec{c}$

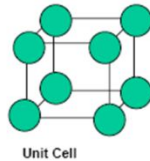
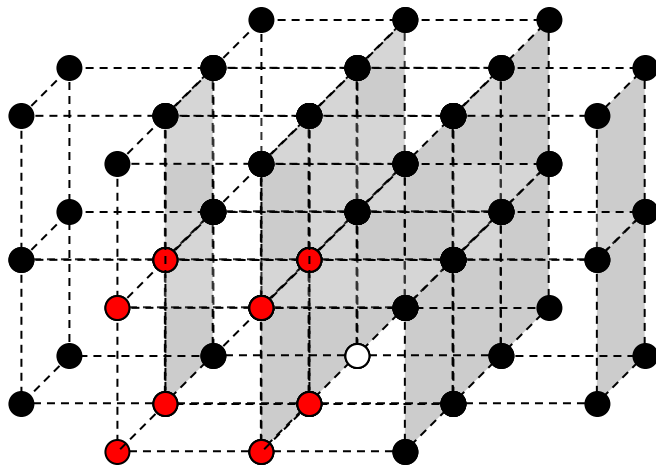
In general: $a \neq b \neq c$

$\alpha \neq \beta \neq \gamma$

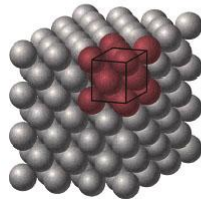
All period unit cells may be described via these vectors and angles.

Unit cell

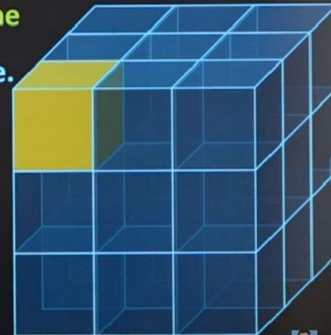
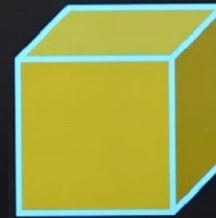
- Unit cell is defined as a fundamental building block of a crystal structure
- Its geometry and atomic positions define the crystal structure.



A **unit cell** is the smallest component of the crystal that reproduces the whole crystal when stacked together with purely translational repetition.



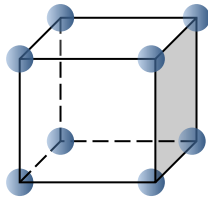
- ❖ The smallest volume of a given crystal structure which is repeated or translated along the three crystallographic axis fills the total lattice structure without leaving any gaps in the lattice is called a unit cell.
- ❖ A unit cell is the smallest repeating portion of a crystal lattice.
- ❖ The properties of lattice and unit cell are same
- ❖ Unit cell is a building block of crystal structure.



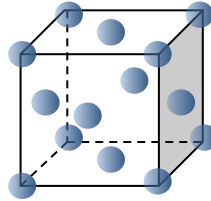
Unit Cells Types

A **unit cell** is the smallest component of the crystal that reproduces the whole crystal when stacked together with purely translational repetition.

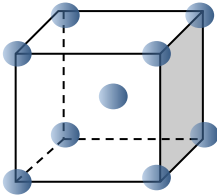
- **Primitive (P)** unit cells contain only a *single lattice point*.
- **Internal (I)** unit cell contains an atom in the *body center*.
- **Face (F)** unit cell contains atoms in the *all faces of the planes* composing the cell.
- **Centered (C)** unit cell contains atoms *centered on the sides* of the unit cell.



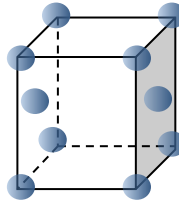
Primitive



Face-Centered



Body-Centered



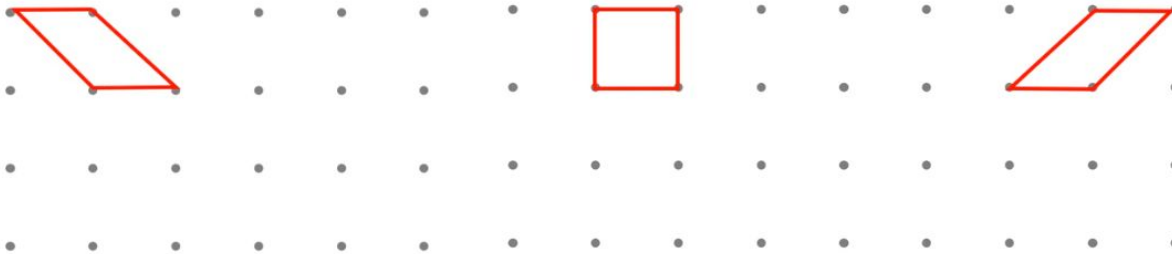
End-Centered

- Sometimes it is convenient to define a non-primitive unit cell to reveal overtly the higher *symmetry*.
- Then, one has to count carefully "how many atoms are in unit cell" (see next).

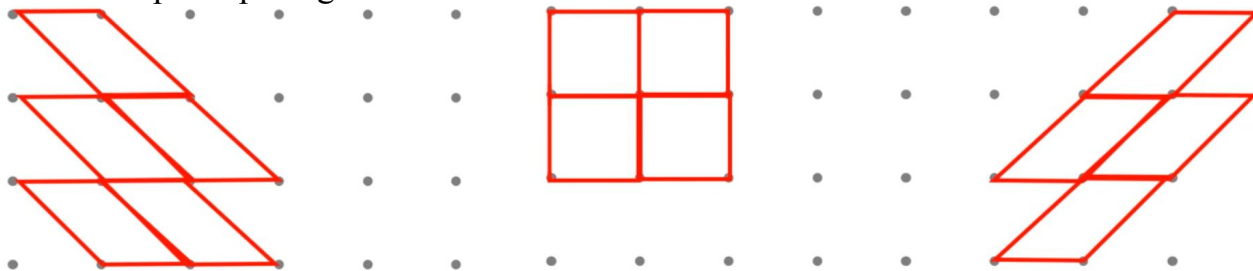
Primitive cell is not unique

- Total contribution of Lattice points should be 1.
- Minimum volume cell
- Repetition of primitive cell should cover entire lattice, without any vacant space or over lapping

Points are the corner of the cell



No overlap no spacing between cells

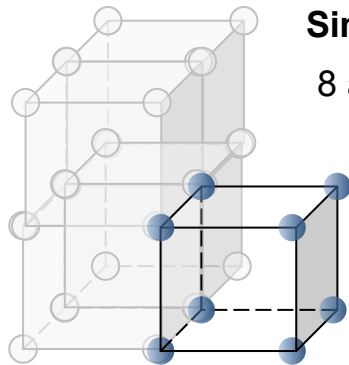


Counting Number of Atoms Per Unit Cell

Counting Atoms in 3D Cells

Atoms in different positions are shared by differing numbers of unit cells.

- **Vertex** atom shared by **8 cells** => **$1/8$** atom per cell.
- **Edge** atom shared by **4 cells** => **$1/4$** atom per cell.
- **Face** atom shared by **2 cells** => **$1/2$** atom per cell.
- **Body** unique to **1 cell** => **1** atom per cell.

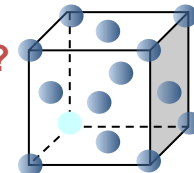
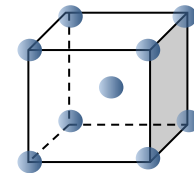


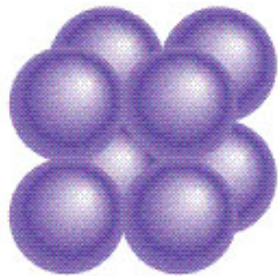
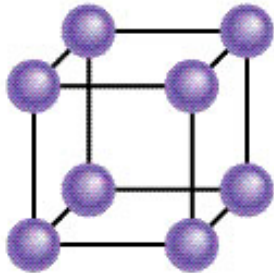
Simple Cubic

8 atoms but shared by 8 unit cells. So,
 $8 \text{ atoms} / 8 \text{ cells} = 1 \text{ atom/unit cell}$

How many atoms/cell for
Body-Centered Cubic?

And, **Face-Centered Cubic?**

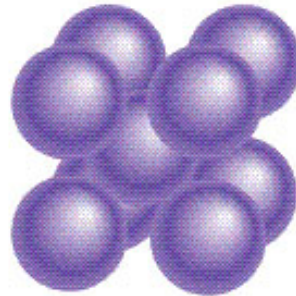
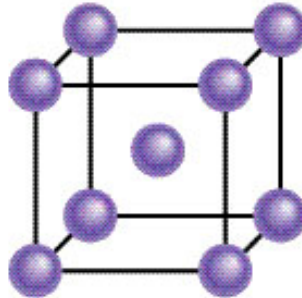




Simple cubic

1 atom/unit cell

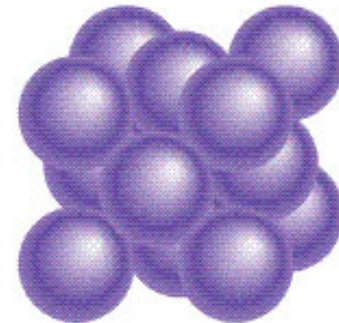
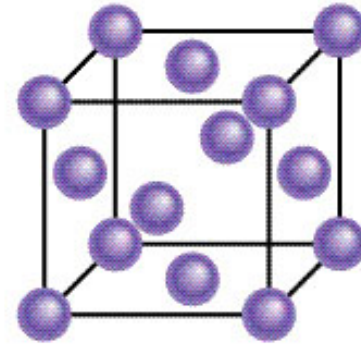
$$(8 \times 1/8 = 1)$$



Body-centered cubic

2 atoms/unit cell

$$(8 \times 1/8 + 1 = 2)$$

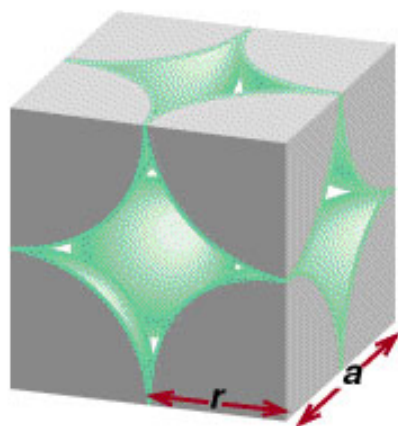


Face-centered cubic

4 atoms/unit cell

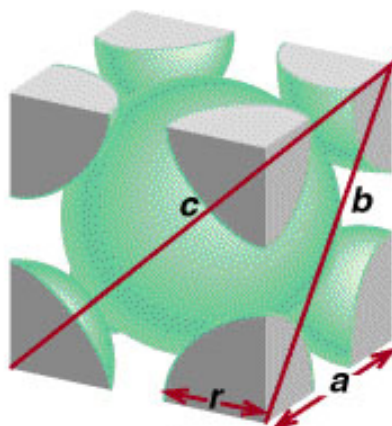
$$(8 \times 1/8 + 6 \times 1/2 = 4)$$

Relationship Between the Atomic Radius and the Edge Length in Three Different Unit Cells



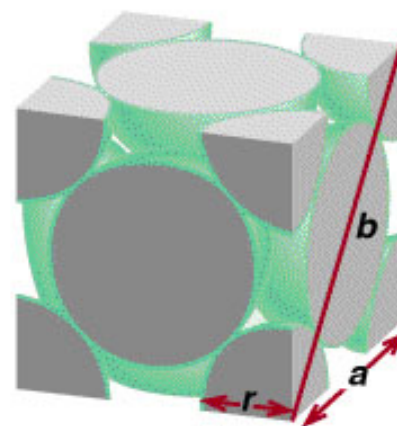
scc

$$a = 2r$$



bcc

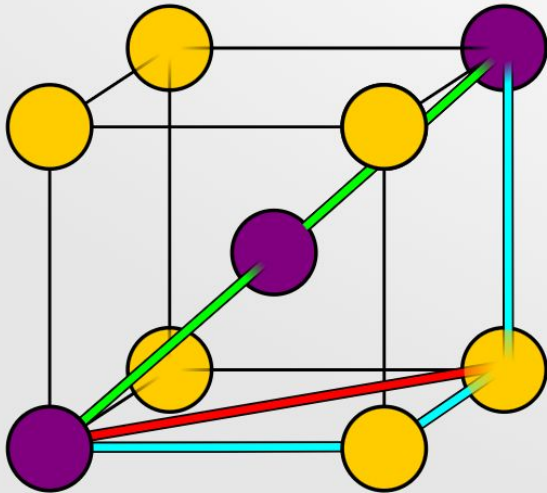
$$\begin{aligned} b^2 &= a^2 + a^2 \\ c^2 &= a^2 + b^2 \\ &= 3a^2 \\ c &= \sqrt{3}a = 4r \\ a &= \frac{4r}{\sqrt{3}} \end{aligned}$$



fcc

$$\begin{aligned} b &= 4r \\ b^2 &= a^2 + a^2 \\ 16r^2 &= 2a^2 \\ a &= \sqrt{8}r \end{aligned}$$

Additional clarification on BCC Lattice

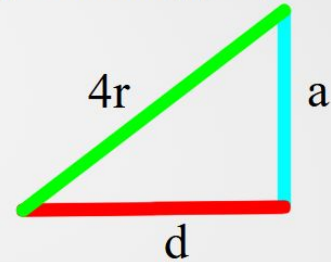


$$\frac{1}{2}2r + \frac{1}{2}2r + 2r = 4r$$

lattice constant a

face diagonal d

2D projection along (110)

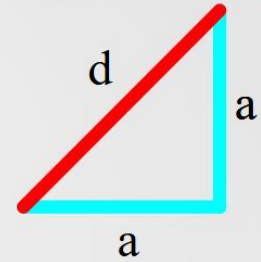


$$a^2 + d^2 = (4r)^2$$

$$3a^2 = (4r)^2$$

$$a = \frac{4}{\sqrt{3}}r$$

2D projection along (001)



$$a^2 + a^2 = d^2$$

$$d^2 = 2a^2$$

Atomic Packing Factor

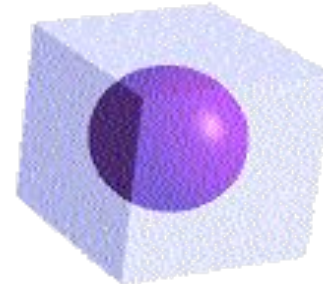
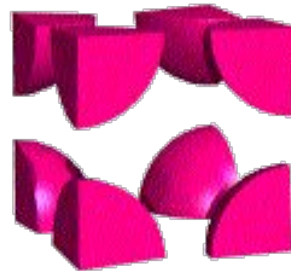
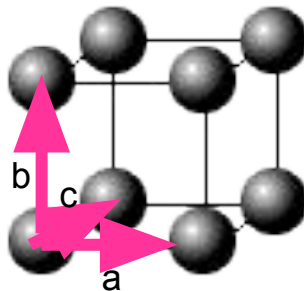
- Atomic Packing Factor (**APF**) is defined as the volume of atoms within the unit cell divided by the volume of the unit cell.

$$\mathbf{APF} = \frac{\text{Volume of Atoms in Unit cell}}{\text{Volume of Unit cell}}$$

1-CUBIC CRYSTAL SYSTEM

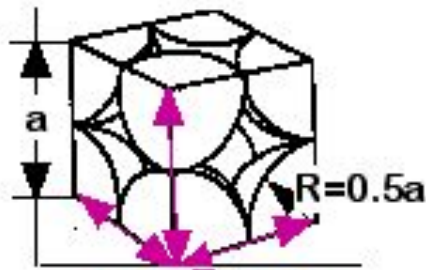
a- Simple Cubic

- Simple Cubic has one lattice point so its primitive cell.
- In the unit cell on the left, the atoms at the corners are cut because only a portion (in this case $1/8$) belongs to that cell. The rest of the atom belongs to neighboring cells.
- Coordination number of simple cubic is 6.



Atomic Packing Factor of Simple Cubic

$$APF = \frac{V_{at}}{V_{uni}}$$



contains $8 \times 1/8 =$
1 atom/unit cell

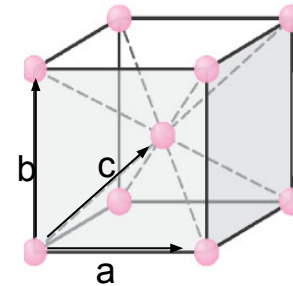
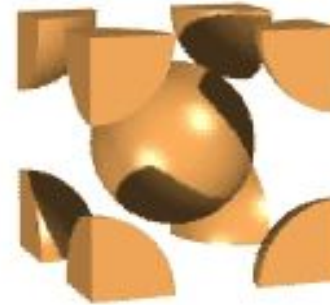
APF = 0.52 for simple cubic

$$APF = \frac{\text{atom/unit cell} \left[1 \frac{4}{3} \pi (0.5a)^3 \right]}{a^3}$$

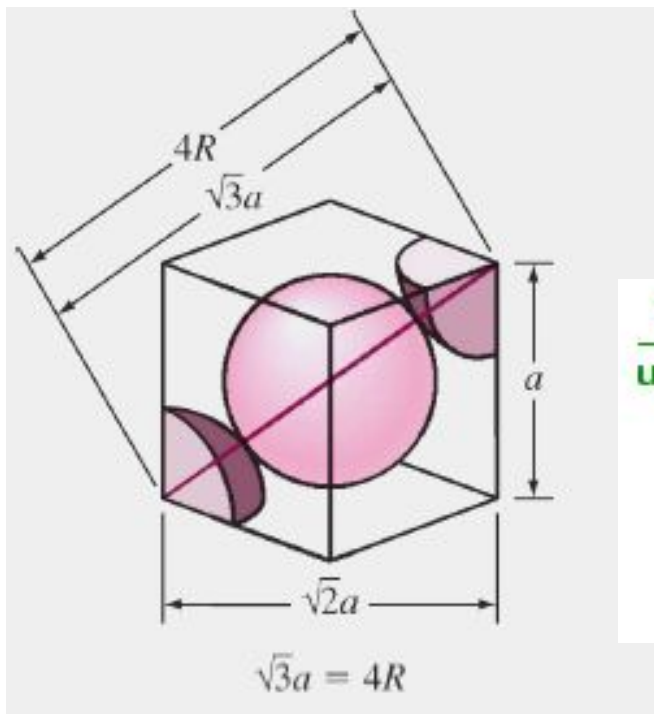
The diagram shows the calculation of the Atomic Packing Factor (APF) for a simple cubic unit cell. The numerator is the total volume of atoms in the unit cell, calculated as the number of atoms per unit cell (1) multiplied by the volume of a single atom ($\frac{4}{3} \pi (0.5a)^3$). The denominator is the volume of the unit cell (a^3). Arrows point from the labels 'atom/unit cell', 'volume/atom', and 'volume/unit cell' to their respective parts in the equation.

b-Body Centered Cubic (BCC)

- BCC has two lattice points so BCC is a non-primitive cell.
- BCC has eight nearest neighbors. Each atom is in contact with its neighbors only along the body-diagonal directions.
- Many metals (Fe, Li, Na..etc), including the alkalis and several transition elements choose the BCC structure.



Atomic Packing Factor of BCC



$$APF_{BCC} = \frac{V_{atoms}}{V_{unit\ cell}} = 0.68$$

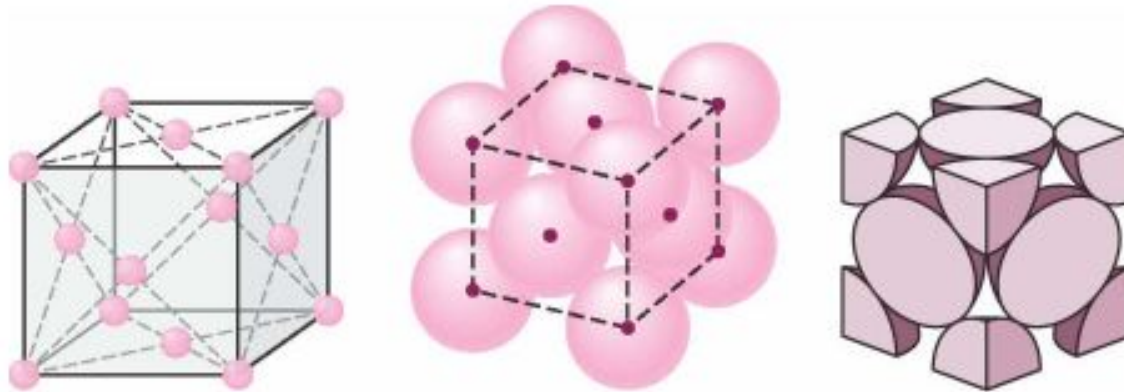
$$APF = \frac{\text{atom unit cell} \left[2 \frac{4}{3} \pi (0.433a)^3 \right]}{a^3}$$

Annotations in the diagram:

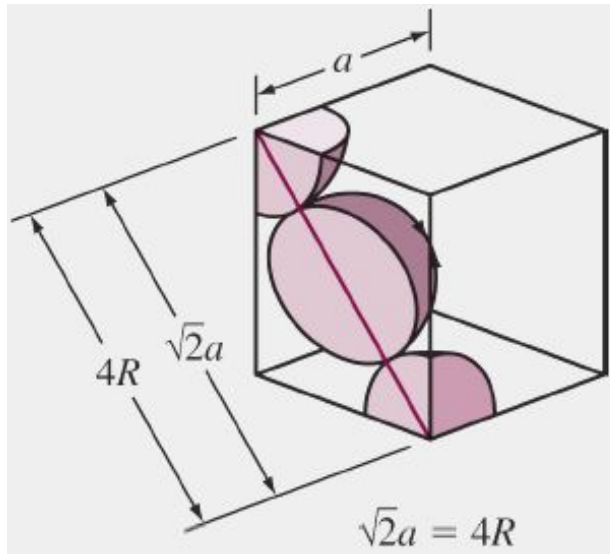
- atom unit cell**: Points to the coefficient 2 in the numerator.
- volume atom**: Points to the term $\frac{4}{3} \pi (0.433a)^3$ in the numerator.
- volume unit cell**: Points to the denominator a^3 .

c- Face Centered Cubic (FCC)

- There are atoms at the corners of the unit cell and at the center of each face.
- Face centered cubic has 4 atoms so its non primitive cell.
- Many of common metals (Cu,Ni,Pb..etc) crystallize in FCC structure.



Atomic Packing Factor of FCC



$$APF_{\boxed{}} = \frac{V_{atoms}}{V_{unit\ cell}} = \boxed{}$$

$$APF = \frac{\overbrace{4 \times \frac{4}{3} \pi (0.353a)^3}^{\substack{\text{atom} \\ \text{unit cell}}}}{\underbrace{a^3}_{\substack{\text{volume} \\ \text{unit cell}}}}$$

THEORETICAL DENSITY, ρ

$$\rho = \frac{nA}{V_c N_A}$$

atoms/unit cell $\rightarrow n$ Atomic weight (g/mol) $\rightarrow A$

Volume/unit cell (cm³/unit cell) $\rightarrow V_c$ Avogadro's number (6.023 x 10²³ atoms/mol) $\rightarrow N_A$

Example: Copper

Data from Table inside front cover of Callister (see next slide):

- crystal structure = FCC: 4 atoms/unit cell
- atomic weight = 63.55 g/mol (1 amu = 1 g/mol)
- atomic radius $R = 0.128$ nm (1 nm = 10⁻⁷cm)
 $V_c = a^3$; For FCC, $a = 4R/\sqrt{2}$; $V_c = 4.75 \times 10^{-23}$ cm³

Result: theoretical $\rho_{Cu} = 8.89$ g/cm³

Compare to actual: $\rho_{Cu} = 8.94$ g/cm³

DENSITIES OF MATERIAL CLASSES

$$\rho_{\text{metals}} > \rho_{\text{ceramics}} > \rho_{\text{polymers}}$$

Metals/Alloys Graphite/Ceramics/Semicond Polymers Composites/fibers

Why?

Metals have...

- close-packing (metallic bonding)
- large atomic mass

Ceramics have...

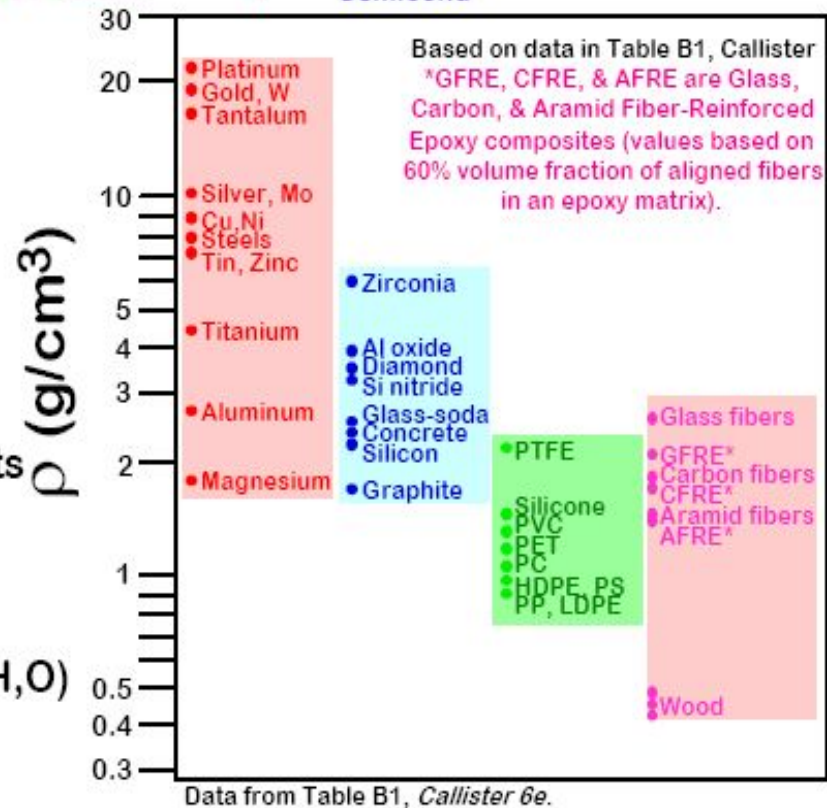
- less dense packing (covalent bonding)
- often lighter elements

Polymers have...

- poor packing (often amorphous)
- lighter elements (C,H,O)

Composites have...

- intermediate values



Types of crystals

- Crystallographers have classified crystalline substances into systems and classes according to the external symmetry of individual well-informed crystals.
- There is a total of 230 possible crystal forms or space groups.
- On the basis of their symmetry, these forms may be grouped into 32 crystal classes or point groups.
- These 32 crystal classes are grouped into **seven crystal system on the basis of the unit cell.**
- These are classified in terms of length of unit cells and the angle of inclination between them.
 - **Cubic**
 - **Triclinic**
 - **Tetragonal**
 - **Trigonal**
 - **Hexagonal**
 - **Orthorhombic**
 - **Monoclinic**

TYPICAL CRYSTAL STRUCTURES

Three dimension (3D)

14 BRAVAIS LATTICES AND THE SEVEN CRYSTAL SYSTEM

- There are only seven different shapes of unit cell which can be stacked together to completely fill all space (in 3 dimensions) without overlapping. This gives the seven crystal systems, in which all crystal structures can be classified.
- Cubic Crystal System (SC, BCC, FCC)
- Hexagonal Crystal System (S)
- Triclinic Crystal System (S)
- Monoclinic Crystal System (S, Base-C)
- Orthorhombic Crystal System (S, Base-C, BC, FC)
- Tetragonal Crystal System (S, BC)
- Trigonal (Rhombohedral) Crystal System (S)

*The Bravais lattice concept was proposed by French mathematician **Auguste Bravais** in **1848**. He described the 14 unique ways that points can be arranged in three-dimensional space to form a regular, repeating pattern, representing the atomic structure of crystals*

Definition of Bravais lattice is given by:

1. Bravais lattice is an infinite array of discrete points, appears exactly same from the point where the lattice is observed.

2. Position vector of any discrete point in the lattice can be written in the form

$$R = n_1 a_1 + n_2 a_2$$

where a_1, a_2 are primitive, linearly independent vectors which spans the whole lattice.

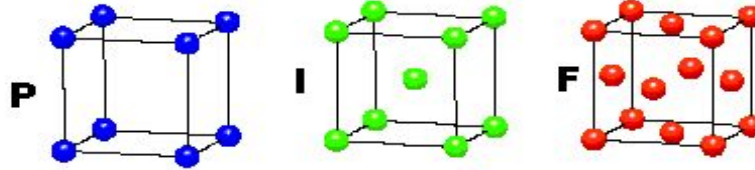
BRAVAIS LATTICES

Combining **7 Crystal Classes** (cubic, tetragonal, orthorhombic, hexagonal, monoclinic, triclinic, trigonal) with **4 unit cell types (P, I, F, C) symmetry** allows for only **14 types of 3-D lattice**.

CUBIC

$$a = b = c$$

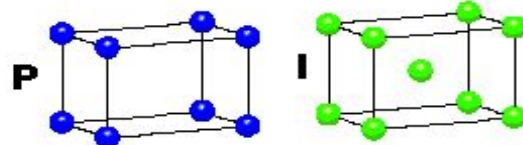
$$\alpha = \beta = \gamma = 90^\circ$$



TETRAGONAL

$$a = b \neq c$$

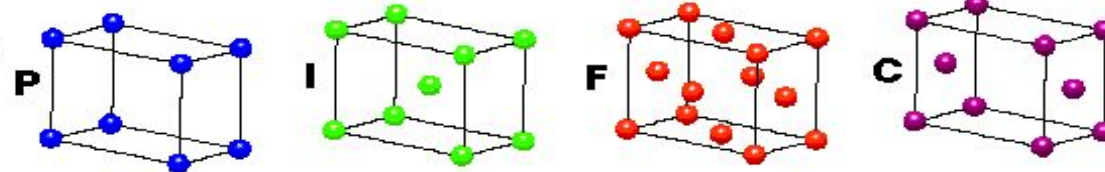
$$\alpha = \beta = \gamma = 90^\circ$$



ORTHORHOMBIC

$$a \neq b \neq c$$

$$\alpha = \beta = \gamma = 90^\circ$$

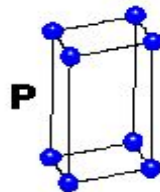


HEXAGONAL

$$a = b \neq c$$

$$\alpha = \beta = 90^\circ$$

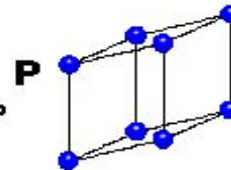
$$\gamma = 120^\circ$$



TRIGONAL

$$a = b = c$$

$$\alpha = \beta = \gamma \neq 90^\circ$$

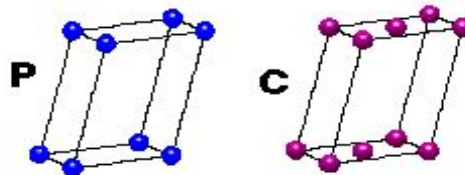


MONOCLINIC

$$a \neq b \neq c$$

$$\alpha = \gamma = 90^\circ$$

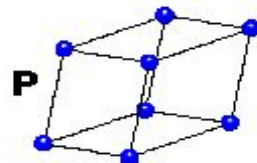
$$\beta \neq 120^\circ$$



TRICLINIC

$$a \neq b \neq c$$

$$\alpha \neq \beta \neq \gamma \neq 90^\circ$$



4 Types of Unit Cell

P = Primitive

I = Body-Centred

F = Face-Centred

C = Side-Centred

+

7 Crystal Classes

→ 14 Bravais Lattices

14 Bravais Lattices divided into 7 Crystal Systems

A Symmetry based concept

'Translation' based concept

	Crystal System	Shape of UC	Bravais Lattices			
			P	I	F	C
1	Cubic	Cube	✓	✓	✓	
2	Tetragonal	Square Prism (general height)	✓	✓		
3	Orthorhombic	Rectangular Prism (general height)	✓	✓	✓	✓
4	Hexagonal	120° Rhombic Prism	✓			
5	Trigonal	Parallelepiped (Equilateral, Equiangular)	✓			
6	Monoclinic	Parallogramic Prism	✓			✓
7	Triclinic	Parallelepiped (general)	✓			

P	Primitive
I	Body Centred
F	Face Centred
C	A/B/C- Centred

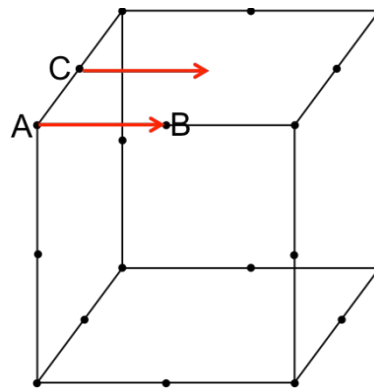
There are only seven different shapes of unit cell which can be stacked together to completely fill all space (in 3 dimensions) without overlapping. This gives the seven crystal systems, in which all crystal structures can be classified

Cubic Crystal System (SC, BCC, FCC)
 Hexagonal Crystal System (S)
 Triclinic Crystal System (S)
 Monoclinic Crystal System (S, Base-C)
 Orthorhombic Crystal System (S, Base-C, BC, FC)
 Tetragonal Crystal System (S, BC)
 Trigonal (Rhombohedral) Crystal System (S)

Why is the edge centred cubic cell not considered a Bravais lattice?

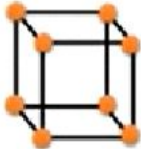
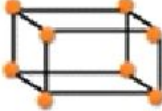
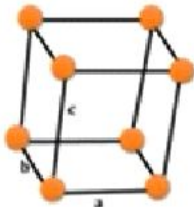
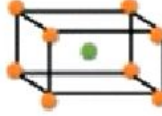
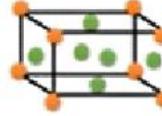
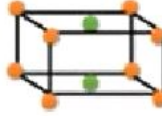
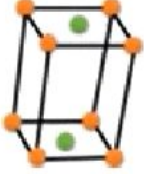
The edge-centered cubic (ECC) cell is not considered a Bravais lattice because it does not meet the criteria for a Bravais lattice. A Bravais lattice is defined as an infinite array of discrete points with the same environment around each point. In the case of the cell, the lattice points are not all equivalent, as the points on the edges of the unit cell are in a different environment compared to the points at the corners. Therefore, the cell does not satisfy the definition of a Bravais lattice.

All points of a lattice should have identical neighbourhood. This means that if I start from a lattice point A and move by a distance d in some direction say x-direction to find another lattice point B then I should find a lattice point C at the same distance in the same direction. This is what is called the translational symmetry of the lattice. We can define a lattice as a set of points having translational symmetry.



Now consider the edge-centered cubic unit cell shown here. It has points at the corners as well as at the centres of each edge. Take a corner point (A) and move horizontally along an edge by distance $a/2$ to find another point (B) (a = cube edge length). Now consider another point C . Moving the same distance $a/2$ along the same direction brings us to a point in the middle of a face. But since it is not a face-centred lattice there is no lattice point there. Thus, C does not have identical neighbourhood to A . Thus the set of points lack translational symmetry and do not form a lattice. Thus, edge centred lattice is not possible in any crystal system.

Why 3D Bravais lattices are of only 14 types

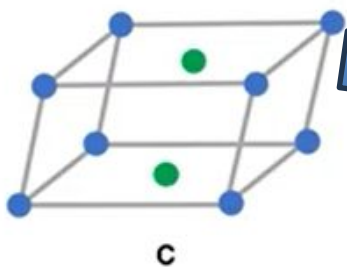
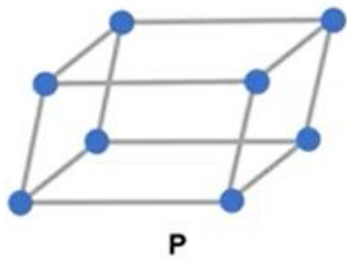
	<u>Cubic Lattice</u>	<u>Tetragonal Lattice</u>	<u>Orthorhombic Lattice</u>	<u>Hexagonal Lattice</u>	<u>Rhombohedral / Trigonal Lattice</u>	<u>Monoclinic Lattice</u>	<u>Triclinic Lattice</u>
<u>Primitive</u>							
<u>Body Centred</u>							
<u>Face Centred</u>							
<u>End Centred</u>							

**Supposed to be $7 \times 4 = 28$
Why not?**

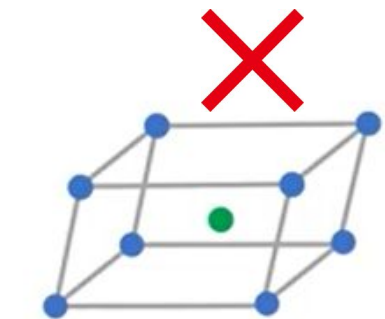
Because there are only 14 different ways to choose nonequivalent basis vectors in 3-space, and using these basis vectors, 14 different special lattice types can be generated

❖ Some of the 28 conceivable lattice types are reductant

Monoclinic Lattice

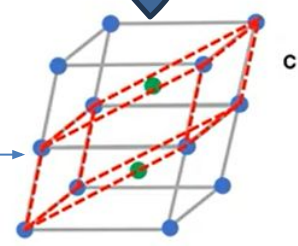


Equivalent



Body centered does not exist

Two-unit cell stuck together
Forming end centered unit cell



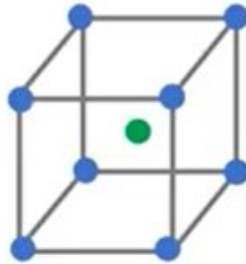
❖ Some of them are not possible due to symmetry reasons

End centered cubic lattice does not exist because it destroys the symmetry (in cubic system)

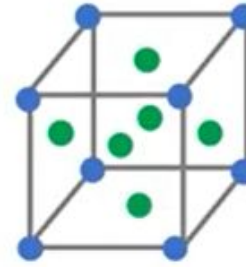
Cubic lattice exist in the following three forms



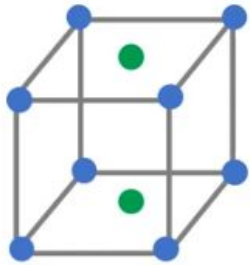
P



I



F



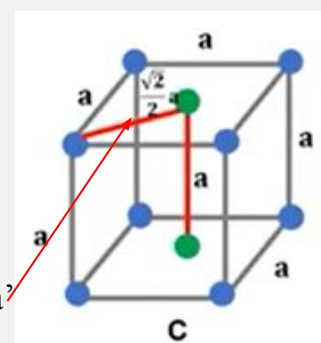
C



It does not exist

Why?

Let's take a close look



Length is not 'a'

So, it is no longer called a cubic lattice

If you add two points at the end of two faces, it will never be cubic anymore.

Why Only 14? (Key Reasons)

1. Redundancy Due to Symmetry

Some centered lattices are **mathematically or geometrically equivalent** to **primitive lattices** in another system. For example:

- A **face-centered tetragonal (FCT)** lattice is **equivalent** to a **body-centered tetragonal (BCT)** lattice when reduced to a primitive cell.
- A **base-centered cubic** lattice is **not unique**, because its symmetry can be described more naturally using a primitive **rhombohedral** or body-centered cell.

So, **some centering types do not introduce new symmetry** and are reducible.

2. Translational Symmetry Restriction

To qualify as a **Bravais lattice**, the arrangement must **remain invariant under translation by lattice vectors**. Some combinations of centering and geometry **break this condition** or lead to a system that already exists in a simpler form.

3. Crystallographic Restrictions

Only certain rotational symmetries are allowed in periodic 3D space:

- Only **1-fold, 2-fold, 3-fold, 4-fold, and 6-fold** rotational symmetries are possible in a periodic lattice.
- This limits the possible lattice configurations that can fill space without gaps or overlaps.

4. Mathematical Group Theory Classification

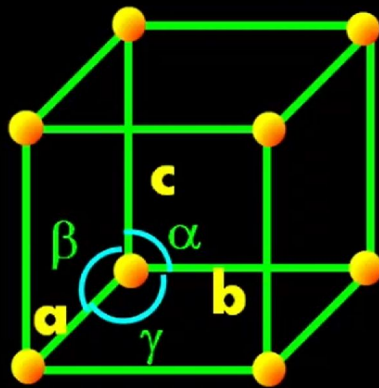
The 14 Bravais lattices emerge as distinct **space-filling lattices** under the constraints of symmetry operations, which are analyzed using **group theory**. These 14 are the only **non-equivalent 3D lattices** under all possible symmetry transformations.

Sr. no.	Crystal System	Bravais Lattice	Lattice Parameters	Example
1	Cubic	S, BC, FC	$a = b = c$ $\alpha = \beta = \gamma = 90^\circ$	Au, Ag, NaCl
2	Tetragonal	S, BC	$a = b \neq c$ $\alpha = \beta = \gamma = 90^\circ$	In, BaTiO ₃ , TiO ₂
3	Orthorhombic	S, BC, FC, BaC	$a \neq b \neq c$ $\alpha = \beta = \gamma = 90^\circ$	U, P, S, Ga
4	Trigonal (Rhombohedral)	S	$a = b = c$ $\alpha = \beta = \gamma \neq 90^\circ$	Hg, Sb, K ₂ Cr ₂ O ₇
5	Monoclinic	S, BaC	$a \neq b \neq c$ $\alpha = \gamma = 90^\circ \neq \beta$	FeSO ₄ , KNO ₂
6	Triclinic	S	$a \neq b \neq c$ $\alpha \neq \beta \neq \gamma \neq 90^\circ$	As, K ₂ S ₂ O ₈
7	Hexagonal	S	$a = b \neq c$ $\alpha = \beta = 90^\circ \neq \gamma = 120^\circ$	C, Zn, Cd

S – Simple ; BC – Body Centered; FC – Face Centered, BaS – Base Centered

1. Cubic

(a) Simple cubic



$$a = b = c$$

$$\alpha = \beta = \gamma = 90^\circ$$

$$\text{C.N.} = 6$$

$$\text{No. of lattice points or atoms per unit cell} = \frac{1}{8} \times 8 = 1$$

Side / Edge of unit cell

$$= a = 2r \text{ i.e. } r = a/2 \quad \underline{r - \text{Radius of atom}}$$

Packing density / Packing fraction of simple cubic = (No. of atoms per unit cell $\times$ Volume of each atom) / Volume of unit cell

$$= (1 \times \frac{4}{3} \pi r^3) / a^3$$

$$= [1 \times \frac{4}{3} \pi (a/2)^3] / a^3$$

$$= [1 \times \frac{4}{3} \pi \cancel{a^3}/8] / \cancel{a^3}$$

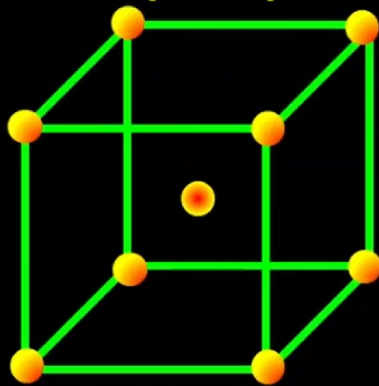
$$= (4 \times \pi) / 3 \times 8 = \pi / 6 = 0.5235$$

$$\text{Volume} = V = a^3 \quad \% \text{ Packing fraction} = 0.5235 \times 100 \approx 52\%$$

$$\text{Number of lattice points} = 8$$

1. Cubic

(b) Body centered Cubic (BCC)



$$a = b = c$$

$$\alpha = \beta = \gamma = 90^\circ$$

$$\text{C.N.} = 8$$

$$\text{Volume} = V = a^3$$

Number of lattice points = 9

$$\begin{aligned} \text{No. of lattice points or atoms per unit cell} \\ = 1/8 \times 8 + 1 = 2 \end{aligned}$$

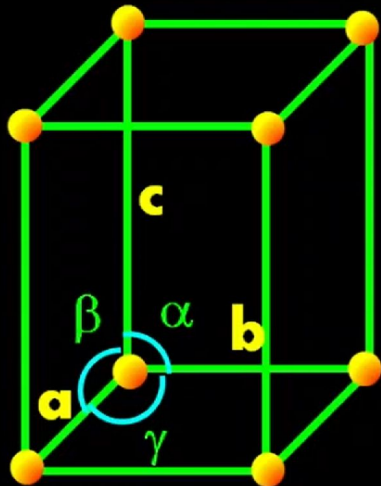
$$\begin{aligned} \text{Side / Edge of unit cell} \\ = a = 4r / \sqrt{3} \text{ i.e. } r = \sqrt{3} a / 4 ; r - \text{Radius of atom} \end{aligned}$$

$$\begin{aligned} \text{Packing density / Packing fraction of BCC} = \\ (\text{No. of atoms per unit cell} \times \text{Volume of each atom}) / \text{Volume of unit cell} \\ = (2 \times 4/3 \pi r^3) / a^3 \\ = [2 \times 4/3 \pi (\sqrt{3} a / 4)^3] / a^3 \\ = [2 \times 4/3 \pi \times 3 \sqrt{3} a^3 / 64] / a^3 \\ = (8 \sqrt{3} \pi) / 64 = \sqrt{3} \pi / 8 = 0.6801 \end{aligned}$$

$$\% \text{ Packing fraction} = 0.6801 \times 100 \approx 68\%$$

2. Tetragonal

(a) Simple tetragonal



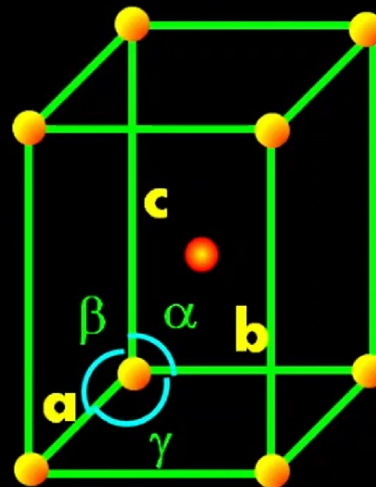
$$a = b \neq c$$

$$\alpha = \beta = \gamma = 90^\circ$$

$$\text{Volume} = V = a^2c$$

$$\text{No. of lattice points or atoms per unit cell} = \frac{1}{8} \times 8 = 1$$

(b) Body centered tetragonal



$$a = b \neq c$$

$$\alpha = \beta = \gamma = 90^\circ$$

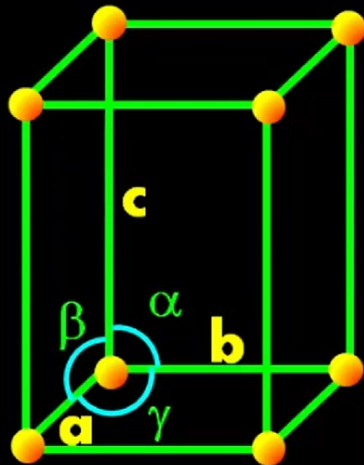
$$\text{Volume} = V = a^2c$$

$$\text{Number of lattice points} = 9$$

$$\text{No. of lattice points or atoms per unit cell} = \frac{1}{8} \times 8 + 1 = 2$$

3. Orthorhombic

(a) Simple orthorhombic



$$a \neq b \neq c$$

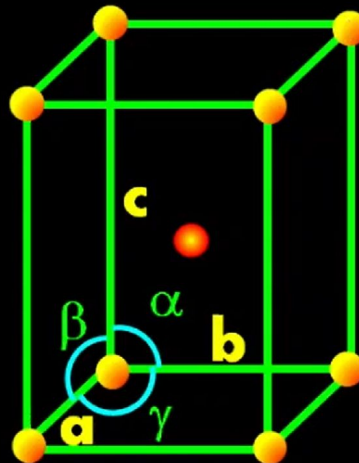
$$\alpha = \beta = \gamma = 90^\circ$$

$$\text{Volume} = V = abc$$

Number of lattice points = 8

No. of lattice points or atoms per unit cell $= 1/8 \times 8 = 1$

(b) Body centered orthorhombic



$$a \neq b \neq c$$

$$\alpha = \beta = \gamma = 90^\circ$$

$$\text{Volume} = V = abc$$

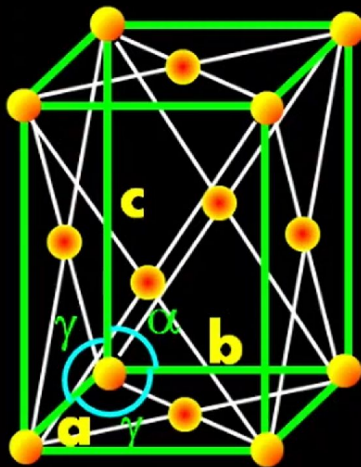
Number of lattice points = 9

No. of lattice points or atoms per unit cell $= 1/8 \times 8 + 1 = 2$

Private Windows
Go to PC settings to activate Windows

3. Orthorhombic

(c) Face centered orthorhombic



$$a \neq b \neq c$$

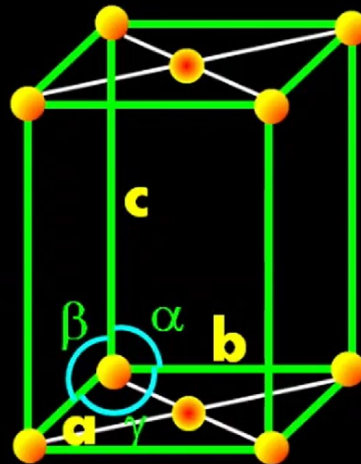
$$\alpha = \beta = \gamma = 90^\circ$$

$$\text{Volume} = V = abc$$

Number of lattice points = 14

$$\text{No. of lattice points or atoms per unit cell} = \frac{1}{8} \times 8 + \frac{1}{2} \times 6 = 1 + 3 = 4$$

(d) Base centered orthorhombic



$$a \neq b \neq c$$

$$\alpha = \beta = \gamma = 90^\circ$$

$$\text{Volume} = V = abc$$

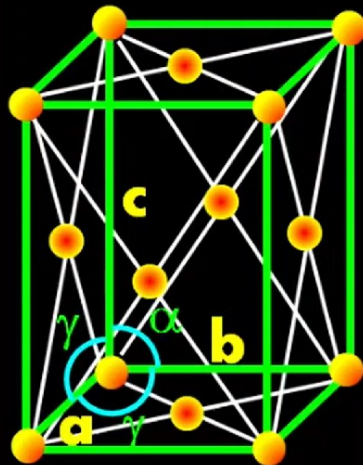
Number of lattice points = 10

$$\text{No. of lattice points or atoms per unit cell} = \frac{1}{8} \times 8 + \frac{1}{2} \times 2 = 1 + 1 = 2$$

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3. Orthorhombic

(c) Face centered orthorhombic



$$a \neq b \neq c$$

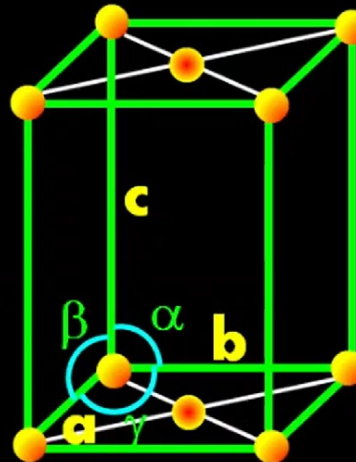
$$\alpha = \beta = \gamma = 90^\circ$$

$$\text{Volume} = V = abc$$

Number of lattice points = 14

No. of lattice points or atoms per unit cell $= 1/8 \times 8 + 1/2 \times 6 = 1 + 3 = 4$

(d) Base centered orthorhombic



$$a \neq b \neq c$$

$$\alpha = \beta = \gamma = 90^\circ$$

$$\text{Volume} = V = abc$$

Number of lattice points = 10

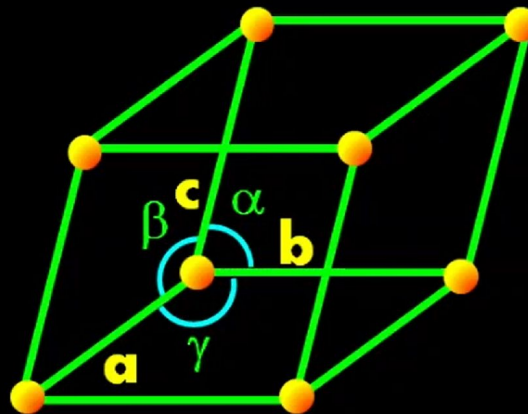
No. of lattice points or atoms per unit cell

$$= 1/8 \times 8 + 1/2 \times 2 = 1 + 1 = 2$$

Activate Windows
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4. Trigonal / Rhombohedral

Simple Trigonal / Rhombohedral



$$a = b = c$$

$$\alpha = \beta = \gamma \neq 90^\circ$$

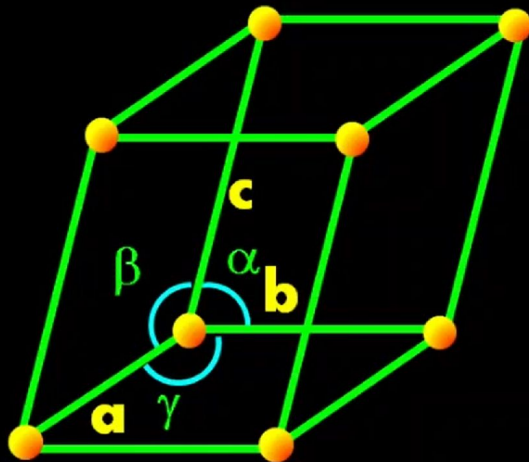
Number of lattice points = 8

No. of lattice points or atoms per unit
cell $= \frac{1}{8} \times 8 = 1$

Activate
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5. Monoclinic

Simple monoclinic



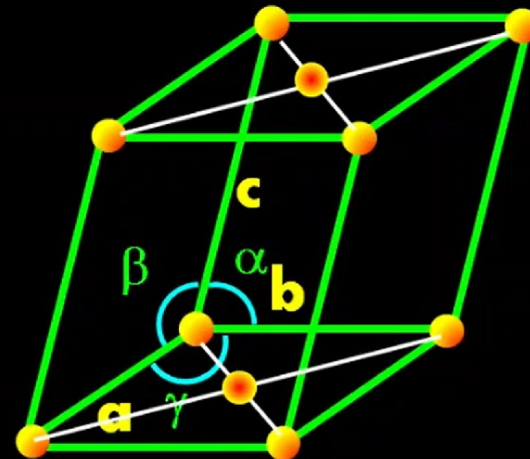
$$a \neq b \neq c$$

$$\alpha = \gamma = 90^\circ \neq \beta$$

Number of lattice points = 8

No. of lattice points or atoms per unit cell = $\frac{1}{8} \times 8 = 1$

Base centered monoclinic



$$a \neq b \neq c$$

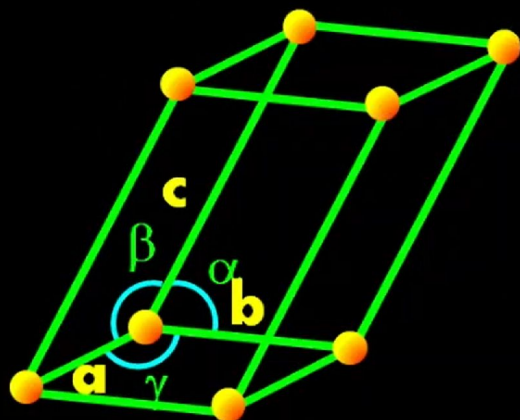
$$\alpha = \gamma = 90^\circ \neq \beta$$

Number of lattice points = 10

No. of lattice points or atoms per unit cell = $\frac{1}{8} \times 8 + \frac{1}{2} \times 2 = 2$

Activate Windows
Go to PC settings

6. Triclinic



$$a \neq b \neq c$$
$$\alpha \neq \beta \neq \gamma \neq 90^\circ$$

Number of lattice points = 8

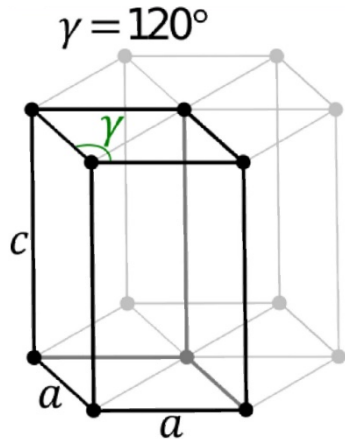
No. of lattice points or atoms per unit cell $= \frac{1}{8} \times 8 = 1$

7. Hexagonal System

The only type of hexagonal Bravais lattice is the *simple hexagonal cell*. It has the following relations between cell sides and angles.

$$a = b \neq c$$

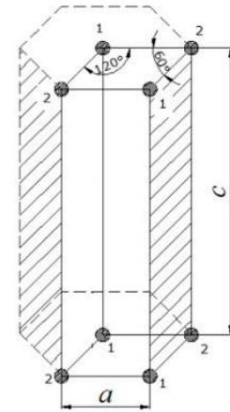
$$\alpha = \beta = 90^\circ \text{ and } \gamma = 120^\circ$$



Zinc oxide and beryllium oxide are made up of simple hexagonal unit cells

Hexagonal primitive unit cell

no. of lattice point in primitive cell
 $= 8 \times 1/8 = 1$ (Bravais lattice)

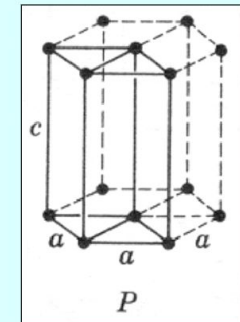


Hexagonal, Hexagonal Lattice

The hexagonal Bravais lattice has only a primitive cell (P). The diagram shows this cell in bold outline. For this lattice: $a = b$ and c is distinct, also $\alpha = \beta = 90^\circ$, $\gamma = 120^\circ$.

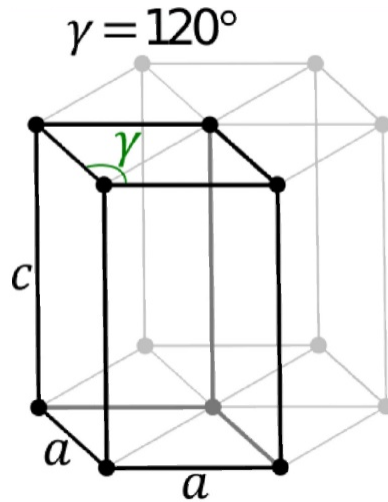
For the primitive cell, partial lattice points are at the cell corners for a total of one lattice point per cell.

The larger "cell" shown in dotted outline displays the hexagonal symmetry of the lattice points, but is not a true unit cell. This construction is used in conjunction with Miller-Bravais indices for this structure. The redundant axis set: a_1, a_2, a_3, c is used for this case.



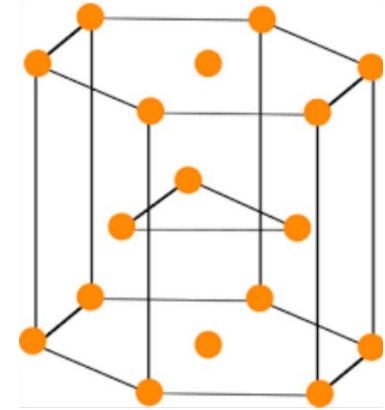
What is the difference between primitive Hexagonal unit cell and Hexagonal closed packing HCP

There are 14 types of Bravais crystal lattices (Primitive + Centered) in 3D. Every solid in this universe is made by the unit cells of these lattices.



Important Note: HCP (Hexagonal Close-Packed) packing and a hexagonal Bravais lattice are different concepts: an HCP structure is a type of crystal structure made of atoms, whereas a hexagonal Bravais lattice is a repeating arrangement of points in space that lacks any atomic basis. HCP is not a Bravais lattice because its atoms don't have identical environments

Hexagonal close packing



The Hexagonal Close-Packed (HCP) *packing* describes the three-dimensional arrangement of atoms or spheres in a crystal. Each sphere in an HCP structure is surrounded by 12 nearest neighbors, giving it a coordination number of 12. In an HCP system, the unit cell is a hexagonal prism, and this specific geometric cell is used to define the overall ABABAB stacking sequence and efficient 74% atomic packing factor characteristic of HCP packing. Examples of metals that adopt this structure include titanium, cobalt, and magnesium

Types of Bravais Lattice in 2D

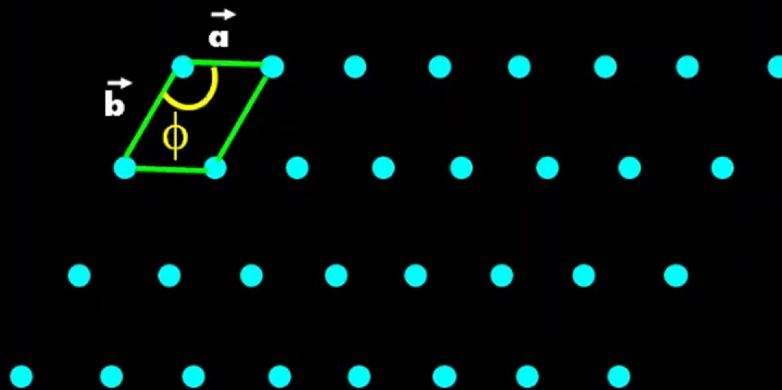
Types of crystal lattice (2D) or Bravais lattices in 2D

Crystals (crystal lattice) are classified into 5 types in 2D.

- 1. Oblique lattice**
- 2. Square lattice**
- 3. Hexagonal lattice**
- 4. Rectangular lattice (Primitive)**
- 5. Centered rectangular lattice**

Types of crystal lattice (2D) or Bravais lattices in 2D

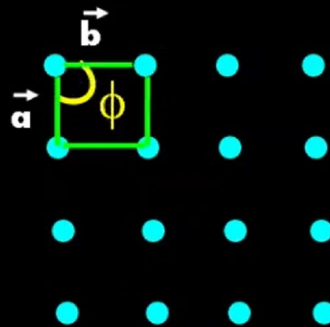
1. Oblique lattice



- Lengths: $|a| \neq |b|$
- Angle: $\theta \neq 90^\circ$

Types of crystal lattice (2D) or Bravais lattices in 2D

2. Square lattice



$$a = b$$

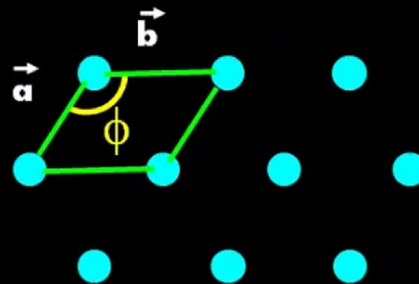
$$\phi = 90^\circ$$

Unit cell – Square

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Types of crystal lattice (2D) or Bravais lattices in 2D

3. Hexagonal lattice

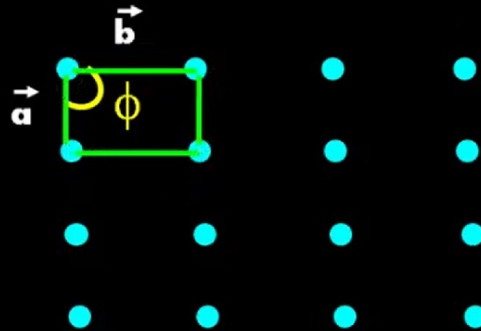


$a = b$ $\phi = 120^\circ$ Unit cell – Rhombus

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Types of crystal lattice (2D) or Bravais lattices in 2D

4. Rectangular lattice (Primitive)

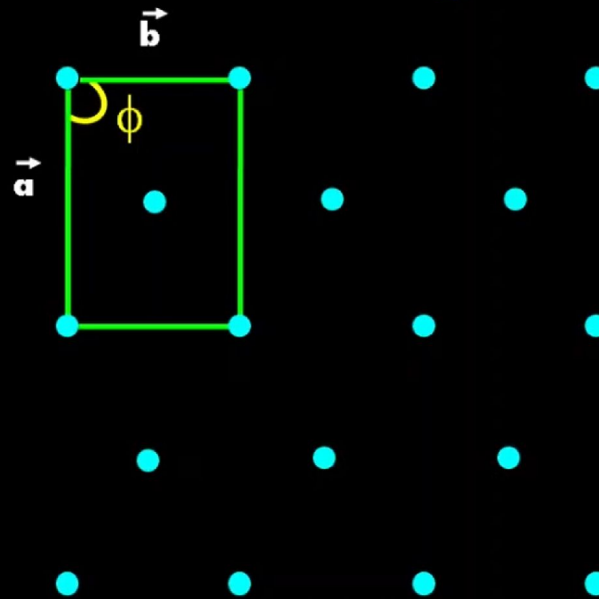


$a \neq b$ $\phi = 90^\circ$ Unit cell – Rectangle

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Types of crystal lattice (2D) or Bravais lattices in 2D

5. Centered Rectangular lattice

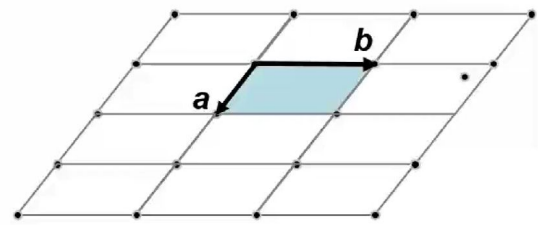


$a \neq b$ ($a > b$) $\phi = 90^\circ$ Unit cell – Rectangle

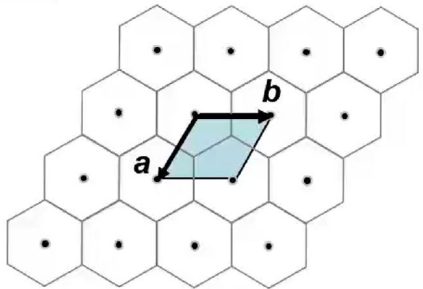
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All 2D Bravais lattices
In a single slide

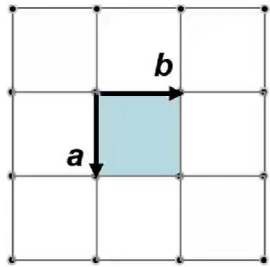
2D Bravais Lattices



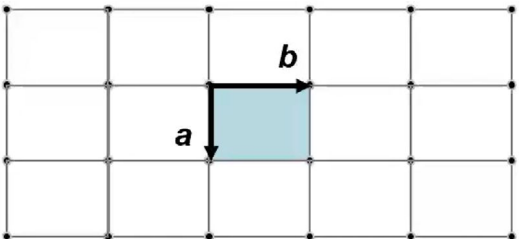
Oblique



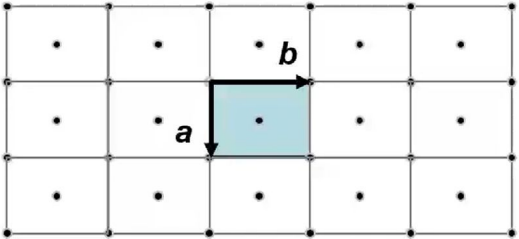
Hexagonal



Square



Primitive Rectangular



Centered Rectangular

Lattice Points and Lattice

- A **lattice point** is just a point in space where you can imagine placing an atom, ion, or a group of atoms (basis).
- These points are arranged in a **periodic, repeating manner** in 1D, 2D, or 3D.
- Mathematically, a lattice point has coordinates given by

$$\vec{R} = n_1 \vec{a}_1 + n_2 \vec{a}_2 + n_3 \vec{a}_3$$

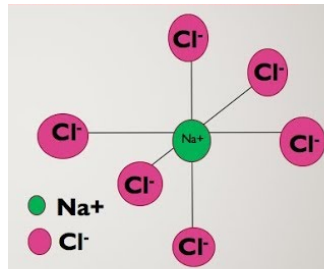
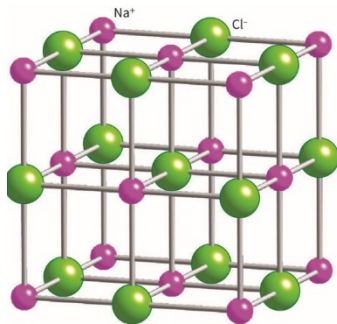
where n_1, n_2, n_3 are integers and $\vec{a}_1, \vec{a}_2, \vec{a}_3$ are the primitive vectors

Lattice

- **General Concept:**
 - Refers to a repeating, periodic arrangement of points in 1D, 2D, or 3D space.
- **Mathematical Definition:**
 - Can be a set of points in any dimension, with its properties defined by a set of primitive translation vectors.
- ***Not Necessarily Equivalent Points:***
 - *A general lattice can contain different types of lattice points, meaning points may not be in identical environments.*

Bravais Lattice

- **Classification of Lattices:** A subset of all possible lattices, specifically those where every point is in the same environment.
- **14 Types:** In three dimensions, there are only 14 distinct Bravais lattices, named after the French mathematician Auguste Bravais.
- **All Points Equivalent:** *Each point within a Bravais lattice has identical neighbors and symmetry.*
- **Crystal Structure Foundation:** A crystal lattice is formed by placing a basis (or motif) at each point of a Bravais lattice.
- **Based on Space Groups:** The 14 Bravais lattices are derived by classifying lattices based on their space group symmetry.
 - **In a Bravais lattice (e.g., simple cubic in 3D)**
 - Pick any lattice point.
 - *Count its nearest neighbors → always the same number, same distance, same orientation.*
Example: In a simple cubic, each point has 6 nearest neighbors at the same distance $a(\pm x, \pm y, \pm z)$ directions)



NaCl, Face centered cubic lattice

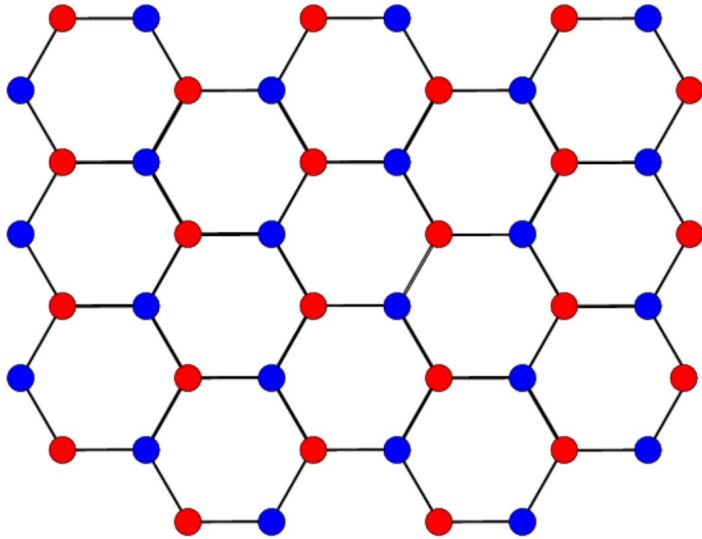
Difference between Lattice and Bravais lattice

A lattice is a general mathematical concept for a periodic set of points in space, while a Bravais lattice is a specific type of lattice where all the lattice points are equivalent and located at the centers of symmetry.

A crystal structure is formed by placing a motif (a group of atoms) at each point of a Bravais lattice, with there being a total of 14 unique 3D Bravais lattices that classify all possible periodic arrangements of points in space.

Non-Bravais crystal lattice

Some of lattice points is non-equivalent when you look at them from different orientation, and the atoms placed at the lattice points are not the same. A non-Bravais lattice is a crystal lattice where not all lattice points are equivalent. Example: (a 2-D honeycomb) The vertices of this 2-D array do not form a Bravais lattice because the orientational relations are not the same when viewed from adjacent points (only if the page is rotated by 180°) each time one moves from one point to the next.



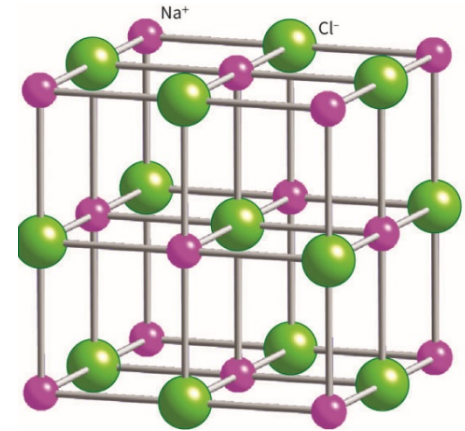
As you can see in the image above, there are two different kinds of lattice sites—coloured red and blue—that are not indistinguishable under translations, and if you were to translate a red site to a blue site, the blue neighbours of the red site would go to the centres of the hexagon where no lattice sites are present.

- *The honeycomb is **not** a Bravais lattice.*
- *Instead, it can be described as a **triangular Bravais lattice** + a **2-atom basis** (two nonequivalent carbon atoms).*

Examples of Non-Bravais Lattices Forming Crystals

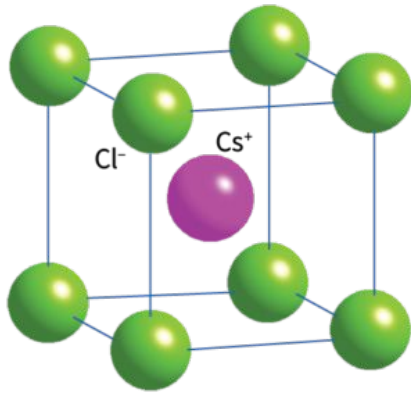
1. NaCl (Sodium Chloride)

- Structure: Each Na^+ is surrounded by 6 Cl^- , and each Cl^- by 6 Na^+ .
- If you look at only the Na or only the Cl atoms $\rightarrow$ they form a **face-centered cubic (fcc) Bravais lattice**.
- But taken together ($\text{Na} + \text{Cl}$), the whole NaCl crystal is **not a Bravais lattice**, because Na and Cl atoms are not identical.



2. CsCl (Cesium Chloride)

- Cs atoms sit at cube corners, Cl at cube center (or vice versa).
- The full structure is a **simple cubic lattice + 2-atom basis** ($\text{Cs} + \text{Cl}$).
- Again, Cs and Cl sites are not equivalent $\rightarrow$ not a Bravais lattice.



What is CsCl Structure?

- CsCl crystal has:
 - Cs atoms at cube corners.
 - Cl atoms at cube center.
- Here, corner atoms and center atoms are different elements, so they are not equivalent.
- Therefore, the structure cannot be described by a single Bravais lattice point with one basis atom

What is BCC (Body-Centered Cubic)?

• In a **true BCC Bravais lattice**,

- **all lattice points are identical.**
- There is **one type of atom** at each lattice point.
- The points can be described as:

$$\vec{R} = n_1 \vec{a}_1 + n_2 \vec{a}_2 + n_3 \vec{a}_3 \text{ with an extra point at } \frac{1}{2}(\vec{a}_1 + \vec{a}_2 + \vec{a}_3)$$

- So, the corner atom and body-center atom are the **same type**, hence **equivalent environments**.

So why is CsCl not BCC?

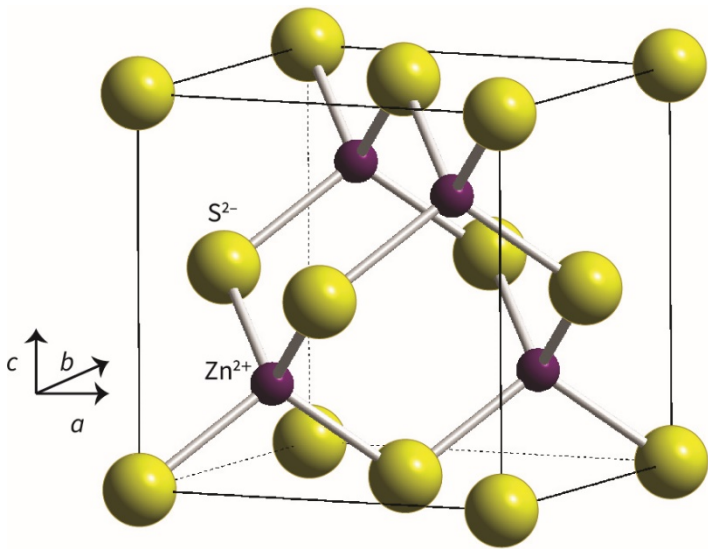
- In **BCC**, corner atom $\equiv$ center atom (same species).
- In **CsCl**, corner atom $\neq$ center atom (different species).
- This breaks the **identical environment condition** required for a Bravais lattice.
- Instead, CsCl is:

Simple *Cubic Bravais lattice* + 2-atom basis (Cs + Cl at 0,0,0 and $\frac{1}{2}, \frac{1}{2}, \frac{1}{2}$).

Examples of Non-Bravais Lattices Forming Crystals

3. ZnS (Zinc Blende)

- Zn and S occupy different positions in an fcc arrangement.
- The two types of atoms form a repeating but **non-equivalent environment**.
- Structure = fcc Bravais lattice + basis (Zn + S).



In **ZnS (zinc blende) structure**, The S^{2-} ions form a face-centered cubic (FCC) lattice, which is a closely packed arrangement. *The Zn^{2+} ions fill half of the tetrahedral voids that exist within the S^{2-} FCC lattice, creating a stable structure where each ion is surrounded by four of the opposite ion.*

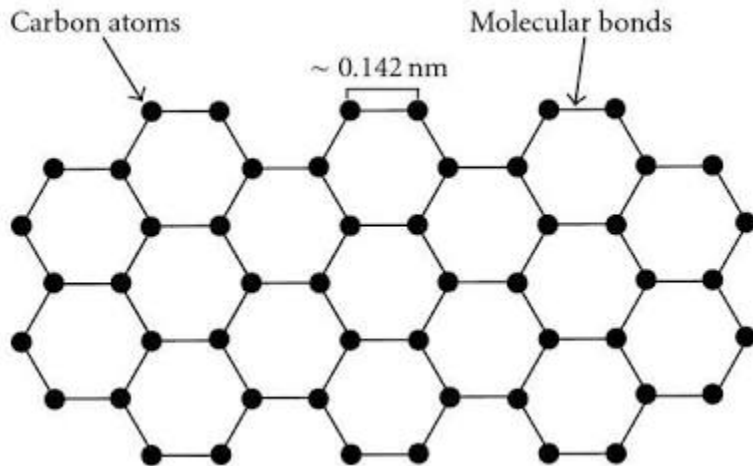
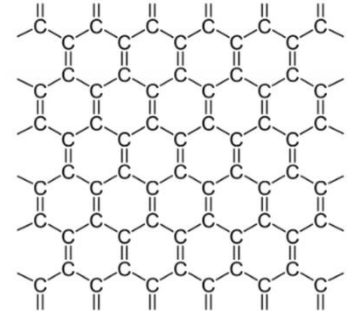
Why ZnS (zinc blende) is not a Bravais lattice? Sulfur (S^{2-}) Ions:

- In a **Bravais lattice**, all lattice points are equivalent.
- In ZnS:
 - Zn atoms occupy one FCC sublattice.
 - S atoms occupy the other FCC sublattice (shifted by $\frac{1}{4}$ along the cube diagonal).
- Since $Zn \neq S$, these two sets of lattice points are **not equivalent**.
- Therefore, ZnS zinc blende = **FCC Bravais lattice + 2-atom basis (Zn, S)**.

Examples of Non-Bravais Lattices Forming Crystals

4. Graphene

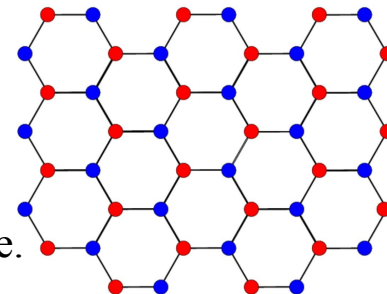
- A 2D honeycomb arrangement of carbon atoms.
- Looks like a hexagonal net, but it's actually a **triangular Bravais lattice + 2-atom basis**.
- The two sublattices (A and B) are not equivalent $\rightarrow$ so honeycomb is **not a Bravais lattice**.



Graphene has a 2D honeycomb structure composed of a single layer of carbon atoms linked by strong sp^2 bonds in a hexagonal lattice, similar to a sheet of chicken wire. This structure gives graphene its unique properties, including exceptional electrical conductivity due to delocalized π electrons, high thermal conductivity, and superior mechanical strength

Why Graphene is Not a Bravais Lattice?

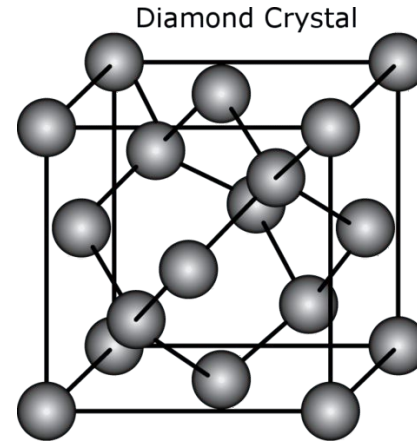
- **Bravais lattice condition:** every lattice point must have an *identical environment*.
- In graphene's honeycomb:
 - There are **two distinct sublattices** (call them A and B).
 - An atom at site **A** has its nearest neighbors all in sublattice **B**.
 - An atom at site **B** has its nearest neighbors all in sublattice **A**.
- So, A and B are **not equivalent points** $\rightarrow$ the whole honeycomb is not a Bravais lattice.



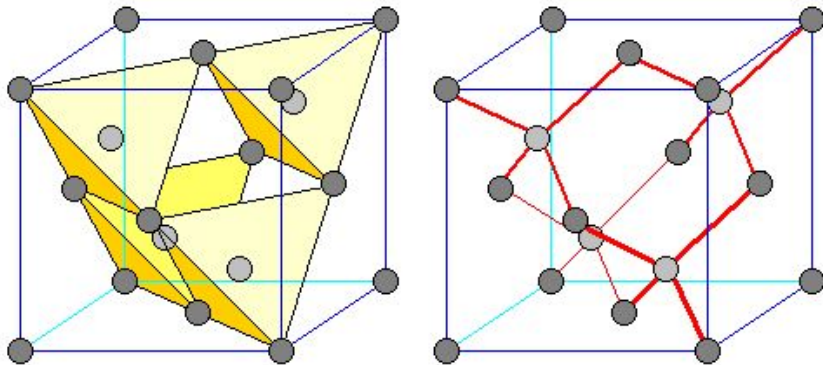
Examples of Non-Bravais Lattices Forming Crystals

5. Diamond

- Carbon atoms in diamond sit in a **tetrahedral arrangement**.
- It can be described as an **fcc Bravais lattice + 2-atom basis**.
- So the full diamond crystal lattice is non-Bravais.

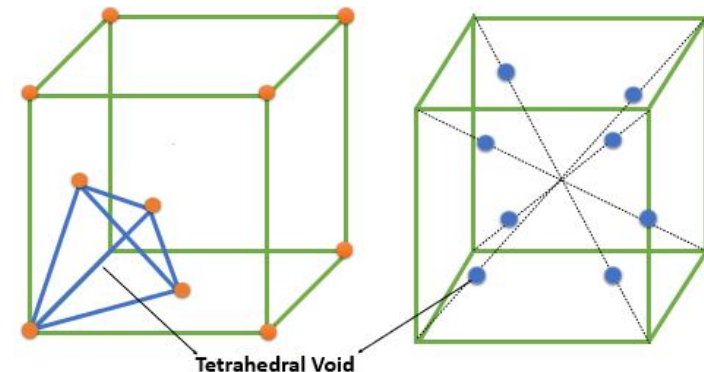


diamond crystal structure



In diamond crystal structure, carbon atoms occupy half of the tetrahedral voids within the face-centered cubic (FCC) lattice formed by other carbon atoms. *There are a total of eight tetrahedral voids in an FCC unit cell, and the diamond structure fills four of these with additional carbon atoms, contributing to a total of 8 carbon atoms per unit cell.*

FCC lattice showing tetrahedral voids = 2×4 (no. of atoms) = 8



Now: Why isn't diamond a Bravais lattice?

Diamond's structure is **not** a Bravais lattice because:

1. Each lattice point is not identical

Diamond has a **face-centered cubic (FCC)** lattice of carbon atoms, but each carbon atom is covalently bonded in a tetrahedral arrangement to four others. The repeating unit is more complex than a single point—it involves a basis.

2. It has a basis

The diamond structure is actually a **face-centered cubic lattice with a two-atom basis**. That is:

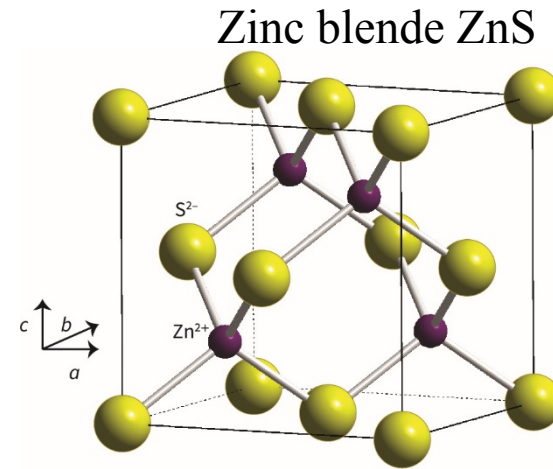
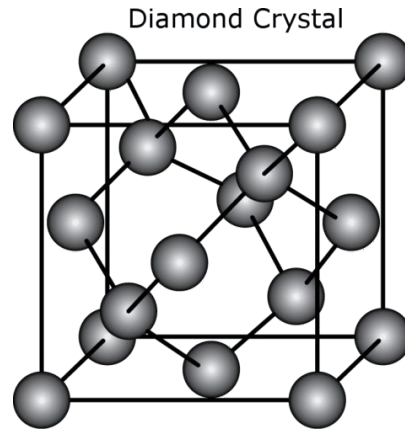
1. One atom is at the origin: $(0, 0, 0)$
2. The second atom is at $(\frac{1}{4}, \frac{1}{4}, \frac{1}{4})$ relative to the origin.

3. These two atoms form the motif that repeats throughout the lattice. Because of this, not every lattice point looks identical in its surroundings unless you include the basis in the description.

4. It's a lattice + basis structure

While the underlying lattice (FCC) **is** a Bravais lattice, the diamond structure as a whole is not because of the two-atom basis that creates a more complex local environment.

Comparison between Zinc Blende ZnS and diamond crystal structure



Key Similarities

- **Tetrahedral Coordination:** In both structures, each atom is surrounded by four neighboring atoms in a tetrahedral arrangement.
- **Interpenetrating FCC Lattices:** Both diamond and zinc blende structures can be viewed as two interpenetrating face-centered cubic (FCC) lattices.
- **Derivation:** The diamond structure is a special case of the zinc blende structure where the two types of atoms are identical, i.e., if all Zn^{2+} and S^{2-} ions in zinc blende are replaced by carbon atoms, it results in the diamond structure.

How They Differ

- **Atom Types:** Diamond consists of only one type of atom (carbon), while the zinc blende structure is composed of two different types of ions (Zn^{2+} and S^{2-}).
- **Lattice Types:** In the diamond structure, the two interpenetrating FCC lattices are made of the same atoms. In the zinc blende structure, the two FCC lattices are made of different ions, which results in

Diamond and the Zinc blende ZnS are same structurally, then why is diamond is the hardest naturally occurring crystals.

1. Bonding strength

•Diamond:

- Pure **covalent bonds** between carbon atoms.
- Carbon-carbon bonds are very strong due to effective orbital overlap.
- Each bond is highly directional and rigid, resulting in a **three-dimensional covalent network**.

•ZnS (zinc blende):

- Bonds between zinc and sulfur have **partial ionic character**.
- The electron sharing is weaker compared to pure covalent bonds.
- This results in a structure that's **less rigid**.

2. Atomic size and packing

- Carbon atoms** are small and tightly packed, allowing for strong overlap of orbitals.
- Zn and S atoms** are larger, leading to less effective overlap and weaker bonds.

3. Bond density

- Diamond has a **higher bond density**, meaning more bonds per unit volume, which makes the structure extremely resistant to deformation.

Crystal Planes

Miller Indices

Planes in a crystal are described by notations called Miller indices

- ❖ Miller indices of a plane, indicated by $h\ k\ l$, are given by the reciprocal of the intercepts of the plane on the three axes.
- ❖ The plane, which intersects X axis at 1 (one lattice parameter) and is parallel to Y and Z axes, has Miller indices $h = 1/1 = 1$, $k = 1/\infty = 0$, $l = 1/\infty = 0$. It is written as $(hkl) = (100)$.
- ❖ Miller indices of some other planes in the cubic system are shown in the figures in the next slide

• General Principles:

- i. If a Miller index is zero, the plane is parallel to that axis.
- ii. The smaller a Miller index, the more nearly parallel the plane is to the axis.
- iii. The larger a Miller index, the more nearly perpendicular a plane is to that axis.
- iv. Multiplying or dividing a Miller index by a constant has no effect on the orientation of the plane
- v. When the integers used in the Miller indices contain more than one digit, the indices must be separated by commas. E.g.: (3,10,13)
- vi. By changing the signs of all the indices of a plane, we obtain a plane located at the same distance on the other side of the origin.

Procedure for finding Miller Indices

- **Step 1** : Determine the **intercepts** of the plane along the axes
- **Step 2** : Determine the **reciprocals** of these numbers.
- **Step 3** : Find the **LCD** and **multiply** each by this *LCD*
- **Step 4** : Write it in paranthesis **in the form (h k l)**.

LCD =least common denominator

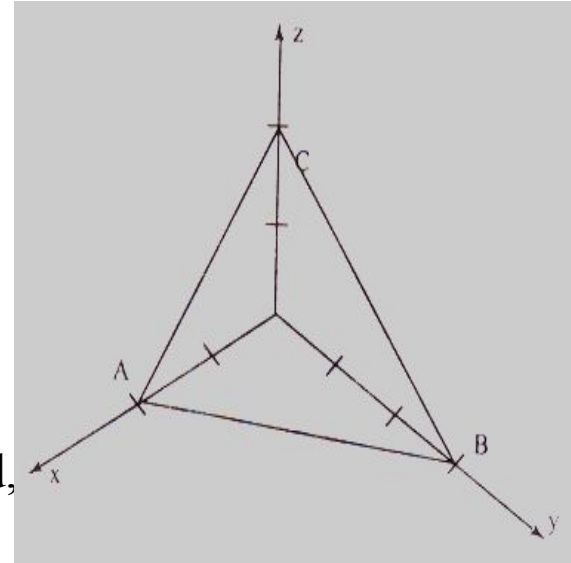
ILLUSTRATION

Step 1 : intercepts - **$2a, 3b$ and $2c$**

Step 2 : reciprocals - **$1/2, 1/3$ and $1/2$** .

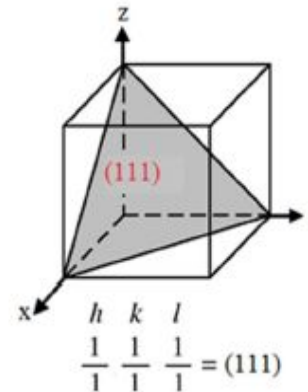
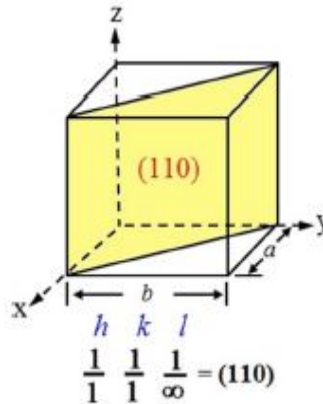
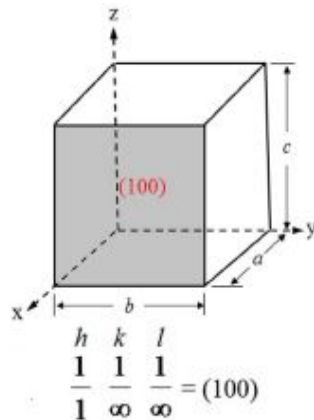
Step 3 : LCD is '6'.

Multiply each reciprocal by lcd,
we get, **$3, 2$ and 3** .



Step 4 : Miller indices for the plane
ABC is **$(3\ 2\ 3)$**

Crystal Planes



To find the Miller Indices of a plane, follow these steps:

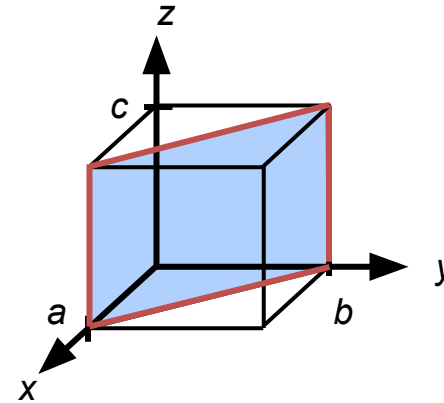
- Determine the intercepts of the plane along the crystal axes
- Take the reciprocals
- Clear fractions
- Reduce to lowest terms and enclose in brackets ()

Ex: Intercepts on a, b, c : $\frac{3}{4}, \frac{1}{2}, \frac{1}{4}$ (h k l) = $(4/3, 2, 4) = (2 \ 3 \ 6)$

Crystallographic Planes

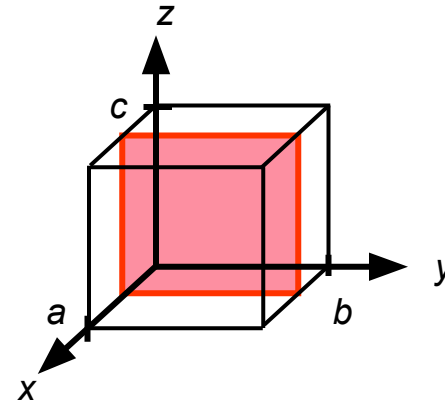
example

	a	b	c
1. Intercepts	1	1	∞
2. Reciprocals	1/1	1/1	1/ ∞
	1	1	0
3. Reduction	1	1	0
4. Miller Indices	(110)		



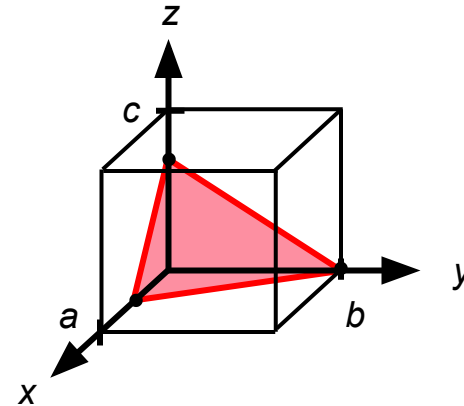
example

	a	b	c
1. Intercepts	1/2	∞	∞
2. Reciprocals	1/1/2	1/ ∞	1/ ∞
	2	0	0
3. Reduction	2	0	0
4. Miller Indices	(100)		



Crystallographic Planes

<u>example</u>	<i>a</i>	<i>b</i>	<i>c</i>
1. Intercepts	1/2	1	3/4
2. Reciprocals	1 1/2	1/1	1 3/4
	2	1	4/3
3. Reduction	6	3	4
4. Miller Indices	(634)		



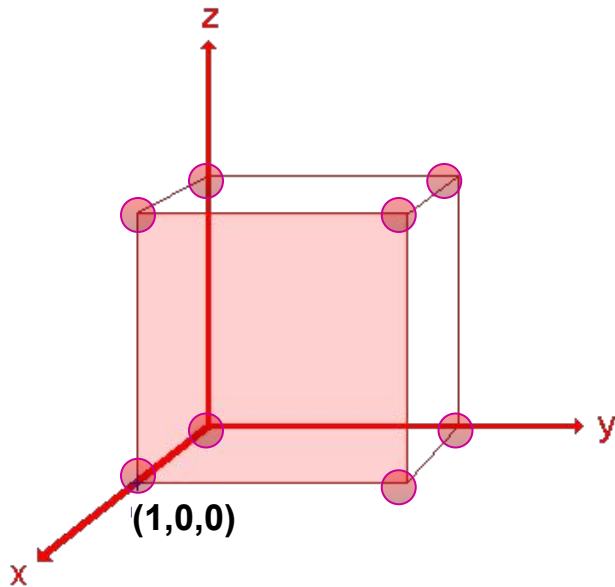
Family of Planes $\{hkl\}$

(second bracket)

(First bracket)

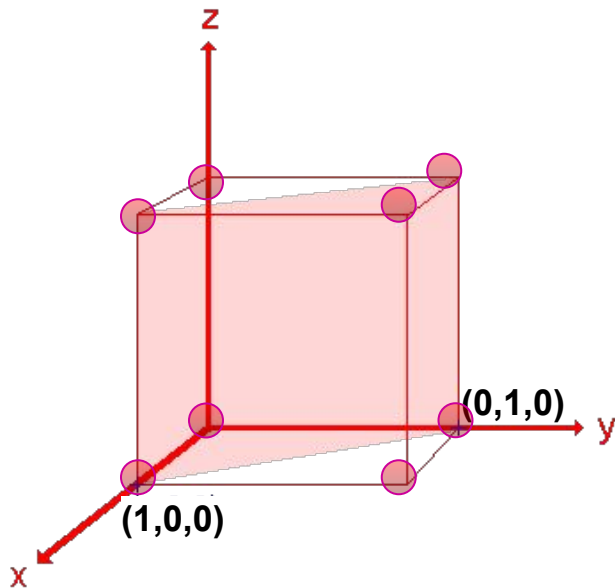
Ex: $\{100\} = (100), (010), (001), (\bar{1}00), (0\bar{1}0), (00\bar{1})$

Example-1



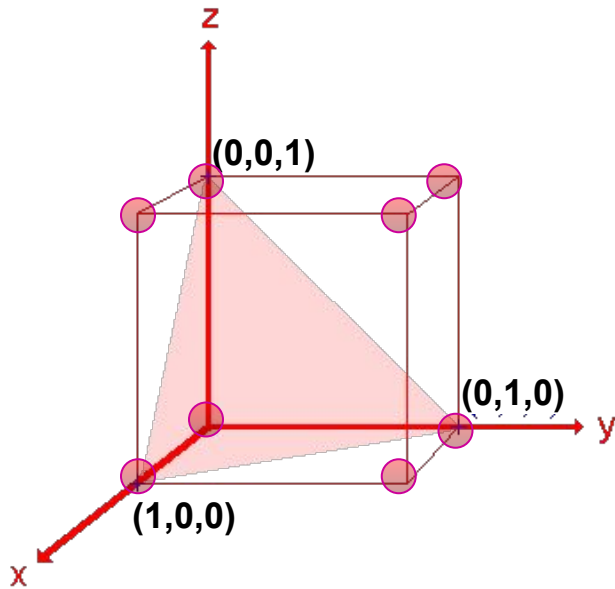
Axis	x	y	z
Intercept points	1	∞	∞
Reciprocals	1/1	1/ ∞	1/ ∞
Smallest Ratio	1	0	0
Miller Indices (100)			

Example-2



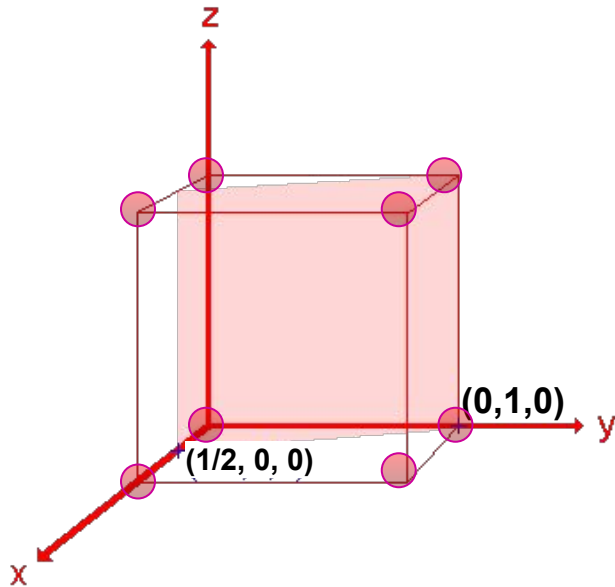
Axis	x	y	z
Intercept points	1	1	∞
Reciprocals	1/1	1/ 1	1/ ∞
Smallest Ratio	1	1	0
Miller Indices (110)			

Example-3



Axis	x	y	z
Intercept points	1	1	1
Reciprocals	1/1	1/ 1	1/ 1
Smallest Ratio	1	1	1
Miller Indices (111)			

Example-4



Axis	x	y	z
Intercept points	$\frac{1}{2}$	1	∞
Reciprocals	$1/(\frac{1}{2})$	$1/1$	$1/\infty$
Smallest Ratio	2	1	0
Miller Indices (210)			

Lattice Sites in Cubic Unit Cell

